UNIVERSAL COLOR GRADING WORKFLOW

DaVinci Resolve Node-Based Scaffolding (Expanded Edition)

A repeatable, professional workflow for narrative, commercial, and documentary projects.

TOOL LOCATION CHEAT SHEET

Tool	Where to Find It
Color Space Transform	Effects Library → OpenFX → ResolveFX Color → Color Space Transform (drag onto node)
Lift / Gamma / Gain / Offset	Color page → Bottom left panel → Color Wheels (icon looks like 4 circles)
Temp / Tint	Color page → Bottom left panel → Color Wheels → Below the Offset wheel, or in Primaries Bars
Custom Curves	Color page → Bottom left panel → Curves (icon looks like an S-curve)
Contrast / Pivot	Color page → Bottom left panel → Primaries Bars → Contrast and Pivot sliders
Luma vs Sat	Color page → Bottom left panel → Curves → Dropdown menu at top → Lum vs Sat
Hue vs Hue	Color page → Bottom left panel → Curves → Dropdown menu at top → Hue vs Hue
Hue vs Sat	Color page → Bottom left panel → Curves → Dropdown menu at top → Hue vs Sat
Hue vs Lum	Color page → Bottom left panel → Curves → Dropdown menu at top → Hue vs Lum
Saturation (global)	Color page → Bottom left panel → Primaries Bars → Sat slider at bottom
Color Boost	Color page → Bottom left panel → Primaries Bars → Below Saturation slider
Midtone Detail	Color page → Primaries Bars (next to Contrast/Pivot) OR Log Wheels bottom bar (next to Color Boost) - same control
Sharpen	Color page → Bottom left panel → Blur (water drop icon) → Sharpen section (H/V sliders)
Power Window (circle)	Color page → Bottom center panel → Window tab → Circle icon
Qualifier	Color page → Bottom center panel → Qualifier tab → HSL controls
Video Limiter	Effects Library → OpenFX → ResolveFX Refinement → Video Limiter (drag onto node)
Color Warper	Color page → Bottom left panel → Color Warper (grid icon)

Tool	Where to Find It
Magic Mask	Color page → Bottom center panel → Magic Mask icon (between Blur and Tracker)
Face Refinement	Effects Library → OpenFX → ResolveFX Refine → Face Refinement (drag onto node)
Tracker	Color page → Bottom center panel → Tracker tab

CORE PHILOSOPHY

World first, style last. Normalize reality before adding taste.

One node = one responsibility. If a node does two jobs, it will fail at both.

Protect highlights and skin early. Everything else depends on that.

Copy structure, adjust values. Never reinvent node trees per shot.

KEYBOARD SHORTCUTS

Action	Shortcut
Go to Color page	Shift + 6
Add Serial Node (after current)	Alt/Option + S
Add Node Before	Alt/Option + B
Add Parallel Node	Alt/Option + P
Add Layer Node	Alt/Option + L
Reset Node (all adjustments)	Cmd/Ctrl + Home
Bypass Node (toggle)	Cmd/Ctrl + D
Bypass All Grades (toggle)	Shift + D
Open Scopes	Cmd/Ctrl + Shift + W
Grab Still	Cmd/Ctrl + Alt + G
Play/Stop	Spacebar
Next Clip	Down Arrow
Previous Clip	Up Arrow

Action	Shortcut
Ripple Grade to All Clips	Right-click node → Ripple Node Changes to Selected Clips
Image Wipe (compare)	Cmd/Ctrl + W
Load clip as reference	Middle-click (or Cmd/Ctrl + click) on any thumbnail

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PRIMARIES VS LOG WHEELS (QUICK EXPLANATION)

You'll see two main wheel modes in Resolve. Here's the difference:

```
┌─────────────────────────────────────────────────────┐
│ TO SWITCH: Click dropdown at top-left of the wheels panel │
│ Options: "Primaries Wheels" / "Primaries Bars" / │
│ "Log" / "HDR" etc. │
└─────────────────────────────────────────────────────┘
```

PRIMARIES (Lift / Gamma / Gain / Offset):

- The classic, default wheels
- Adjustments OVERLAP - Lift affects Gamma a little, Gamma affects Gain a little
- More natural, organic feel to adjustments
- Good for general correction work

LOG (Shadow / Midtone / Highlight / Offset):

- More ISOLATED adjustments - each wheel affects only its range
- You can customize the ranges (Low Range, High Range sliders)
- Designed for log-encoded footage
- Good for targeted, surgical adjustments

Which to use?

Situation	Recommended
General correction, starting out	Primaries
Working with LOG footage	Log wheels often feel more natural
Need to affect ONLY shadows	Log (more isolated)
Want organic, film-like response	Primaries
Need precise control over ranges	Log (adjustable ranges)

Important: Both Primaries and Log adjustments work INDEPENDENTLY. If you adjust Primaries, then switch to Log and adjust, BOTH are applied. Reset one doesn't reset the other.

The sliders around the wheels (Contrast, Mid/Detail, Color Boost, etc.) are shared - they're the same controls regardless of which wheel mode you're in.

PROJECT-LEVEL COLOR MANAGEMENT (ALTERNATIVE TO N1)

You can handle input color space at the project/timeline level instead of using a node. This is often cleaner for single-camera projects.

```
┌──────────────────────────────────────────────────────────┐
│ LOCATION: File → Project Settings → Color Management    │
│ Or: Click the gear icon (bottom-right of screen)        │
└──────────────────────────────────────────────────────────┘
```

Setup for color-managed workflow:

1. **Color Science:** Set to "DaVinci YRGB Color Managed"
2. **Input Color Space:** Set your camera's color space (e.g., S-Gamut3.Cine / S-Log3)
3. **Timeline Color Space:** Set to DaVinci Wide Gamut / Intermediate
4. **Output Color Space:** Set to your delivery (usually Rec.709 for web)

This applies the transform to everything automatically - you can skip N1 entirely.

For mixed cameras on the same timeline:

1. Right-click a clip in the Media Pool
2. Select **Input Color Space** → Choose that camera's specific space
3. Resolve handles the conversion per-clip automatically

When to use which approach:

Approach	Use When
Project-level color management	Single camera, consistent footage
Per-clip Input Color Space	Mixed cameras, B-roll from different sources
N1 node (Color Space Transform)	Need per-shot control, want to see before/after, or learning

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HOW TO READ SCOPES (CHECKING LUMA VALUES)

When the document says "skin should be at 40-55%," here's how to actually check that.

```
┌─────────────────────────────────────────────────────────────┐
│ OPEN SCOPES: Cmd/Ctrl + Shift + W                            │
│ Or: Workspace menu → Video Scopes                            │
│ CHANGE SCOPE TYPE: Right-click on scope → Select type        │
└─────────────────────────────────────────────────────────────┘
```

Method 1: Waveform + Visual Correlation (Everyday Method)

The waveform displays your image laid out left-to-right, with brightness on the vertical axis:

- **Bottom (0%)** = Pure black
- **Top (100%)** = Pure white
- **Middle (50%)** = Midtones

How to read it:

1. Look at where something is positioned horizontally in your image (e.g., face on the left side)
2. Look at the same horizontal position on the waveform
3. See where the "blob" of pixels sits vertically - that's your brightness percentage

Method 2: Qualifier Picker

1. Go to **Qualifier tab** (bottom center panel)
2. Click the **eyedropper** tool
3. Click on the area you want to measure (e.g., someone's cheek)
4. The selected area highlights on the waveform, showing you exactly where it falls

Method 3: Data Burn-In (Exact Numbers)

For precise values when you need actual numbers:

1. **Workspace → Data Burn-In**
2. Enable **"Cursor Position RGB"** or similar option
3. Now when you hover over the image, exact values display on screen
4. Hover over skin → read the Y (luma) value directly

Note: This is overkill for normal grading. Most colorists eyeball it using the waveform. You'll develop an intuition for "that blob is around 45%" after a while.

Quick Visual Reference

What You're Checking	Where It Should Sit on Waveform
Darkest shadows	Just touching the bottom (0-5%)
Dark skin tones	Lower-middle area (35-45%)

What You're Checking	Where It Should Sit on Waveform
Medium skin tones	Middle area (45-55%)
Light skin tones	Upper-middle area (55-65%)
White objects	Upper area (80-90%)
Brightest highlights	Near top but not clipping (90-98%)

SHOT COMPARISON AND MATCHING

When matching shots, you need to compare your current clip against a reference. Here's how.

```
GRAB STILL: Right-click thumbnail → Grab Still
          Or: Cmd/Ctrl + Alt + G

COMPARE: Double-click a still in Gallery to load it
         Then press Cmd/Ctrl + W for wipe view
```

Setting Up a Reference Still

1. Grade your **hero shot** (the shot you want everything else to match)
2. **Grab a still:** Right-click the thumbnail → Grab Still (or Cmd/Ctrl + Alt + G)
3. The still appears in the **Gallery** (top-left panel in Color page)

Comparing Against Your Reference

1. Move to the next shot you want to grade
2. In the Gallery, **double-click your reference still** - this loads it for comparison
3. Press **Cmd/Ctrl + W** to enable the wipe/split view
4. Drag the wipe line across to compare the two images side by side

Wipe Modes

Click the dropdown next to the wipe button in the viewer to change modes:

Mode	What It Does	Use For
Wipe	Diagonal/straight line split you can drag	General comparison, seeing both at once
Split Screen	Side by side, 50/50	Direct comparison without overlap
Blend	Opacity mix between the two	Checking if exposures match
Difference	Shows pixel differences (black = identical)	Technical matching, finding discrepancies

Comparing Against Another Clip (Not a Still)

If you want to compare against a clip you haven't grabbed as a still:

1. **Hover over any clip** in the thumbnail timeline at the bottom of the Color page
2. **Middle-click** (scroll wheel click) or **Cmd/Ctrl + click** to load it as the reference
3. Press **Cmd/Ctrl + W** to wipe/compare

Matching Workflow Tips

- **Match skin tones first** - If faces match, most other things will fall into place
- **Look at the waveform while comparing** - Both images should have similar waveform shapes
- **Match brightness before color** - Get exposure right, then worry about color balance
- **Use the same area of frame** - Compare face-to-face, not face-to-sky

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HORIZONTAL WORKFLOW (HOW TO WORK THROUGH SHOTS)

Critical concept: Work HORIZONTALLY across shots, not VERTICALLY through nodes.

What This Means

CORRECT - Horizontal workflow:

```
N2 on shot 1 → N2 on shot 2 → N2 on shot 3 → ... → N2 DONE on all shots
N3 on shot 1 → N3 on shot 2 → N3 on shot 3 → ... → N3 DONE on all shots
N4 on shot 1 → N4 on shot 2 → N4 on shot 3 → ... → N4 DONE on all shots
...continue through N10
```

WRONG - Vertical workflow:

```
Shot 1: N2 → N3 → N4 → N5 → N6 → N7 → N8 → N9 → N10 (fully finished)
Shot 2: N2 → N3 → N4 → N5 → N6 → N7 → N8 → N9 → N10 (start from scratch)
Shot 3: ... ← This is a nightmare, don't do this
```

Why Horizontal Works Better

Reason	Explanation
Easier matching	When you're doing N2 on shot 5, shots 1-4 are at the same stage - you're comparing apples to apples
Consistent eye	You're making the same type of decision repeatedly, so you stay calibrated
Catch mistakes early	If N2 is wrong on shot 3, you'll notice immediately when you do shot 4
Look stays consistent	The creative look (N9) gets applied to already-matched footage

The Full Process

Phase 0: Setup

1. Build your 10-node structure on the hero shot
2. Save as PowerGrade (see below)
3. Apply the PowerGrade to all clips in the scene

Phase 1: Physics (N1-N4) - All shots 4. Complete N2 (balance) on ALL shots - use wipe to compare each to your hero 5. Complete N3 (contrast) on ALL shots 6. Complete N4 (highlight safety) on ALL shots

Phase 2: Perception (N5-N7) - All shots 7. Complete N5 (saturation) on ALL shots 8. Complete N6 (hue sculpt) on ALL shots 9. Complete N7 (texture) on ALL shots

Phase 3: Taste (N8-N10) - All shots 10. Complete N8 (vignette) on shots that need it 11. Complete N9 (look) on ALL shots 12. Verify N10 (output) on ALL shots

Working in Scene Batches

For longer projects, work scene by scene:

1. Scene 1: N2 all shots → N3 all shots → ... → N10 all shots
2. Scene 2: N2 all shots → N3 all shots → ... → N10 all shots
3. And so on...

This keeps you focused and ensures each scene is internally consistent before moving on.

POWERGRADE: SAVING AND REUSING NODE STRUCTURES

A PowerGrade is a saved node structure that works across ALL projects (unlike regular stills which are project-specific).

PowerGrade vs Regular Still

Still	PowerGrade
Saved in current project's Gallery	Saved to disk as a file
Only accessible in this project	Accessible from ANY project
Lost if project corrupts	Survives independently
Good for shot references	Good for templates and reusable structures

Creating a PowerGrade

1. Build your node tree (the 10-node structure, with or without values)
2. **Grab a still:** Right-click thumbnail → Grab Still (or Cmd/Ctrl + Alt + G)
3. Go to **Gallery** (top-left panel in Color page)
4. Find your still → **Right-click** → **Export** → **Export as PowerGrade**
5. Save the .dpx file somewhere you'll remember (create a "LUTs and Grades" folder)

Using a PowerGrade

1. Open any project
2. In the Gallery → **Right-click** → **Import** → **Import PowerGrade**
3. Navigate to your saved .dpx file
4. Select the clips you want to apply it to
5. **Right-click the PowerGrade** → **Apply Grade**

PowerGrade Tips

- **Create a "template" PowerGrade** with all 10 nodes labeled but minimal adjustments
- **Create camera-specific PowerGrades** with N1 already set for each camera you use
- **Create look PowerGrades** with your favorite N9 looks for quick application
- **Keep PowerGrades organized** in folders by project type or camera

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MASTER NODE STRUCTURE

How to create the 10-node tree:

1. Select your clip in the Color page

2. Press Alt/Option + S to add a serial node (repeat 9 times for 10 total)
3. Right-click each node → "Node Label" → Name them N1, N2, etc.
4. Or: Right-click in the node graph → Add Node → Serial

```
N1  Input Transform / Camera Normalization
N2  Primary Balance (Exposure + White Balance)
N3  Contrast Shaping
N4  Highlight Safety
N5  Saturation & Color Density
N6  Hue Sculpt (Selective Color Control)
N7  Texture Control
N8  (Optional) Vignette / Eye Guidance
N9  Creative Look
N10 Output Trim / Legal Safe
```

Mental model:

- N1–N4 = Physics (making the image technically correct)
- N5–N7 = Perception (making it feel right to human eyes)
- N8–N9 = Taste (artistic choices)
- N10 = Insurance (delivery safety)

NODE-BY-NODE DETAILED BREAKDOWN

N1 — INPUT TRANSFORM / NORMALIZATION

Purpose: Bring footage into a predictable working color space so every adjustment you make afterward behaves consistently.

```
┌─────────────────────────────────────────────────┐
│ TOOL: Color Space Transform                     │
│ LOCATION: Effects Library → OpenFX → ResolveFX Color → │
│         Color Space Transform                   │
│ HOW: Drag and drop onto the node                │
└─────────────────────────────────────────────────┘
```

What This Node Actually Does

When you shoot in LOG (like S-Log3, C-Log, V-Log), your camera captures a "flat" image with lots of latitude for grading. But LOG footage looks washed out and gray - that's intentional. The camera is storing maximum dynamic range, not a pretty picture.

This node translates that LOG data into a standardized working space (DaVinci Wide Gamut / Intermediate) so that when you push the color wheels, the image responds predictably.

Think of it like this: Different cameras speak different languages. This node is the translator that converts everything to one common language before you start working.

When You Need This Node

USE IT when:

- You shot LOG footage (S-Log, C-Log, V-Log, BRAW, ProRes RAW, etc.)
- You're mixing different cameras on the same timeline
- Phone footage is behaving weirdly (too contrasty, colors shifting unexpectedly)
- You're working in a color-managed workflow (ACES, DaVinci Color Management)

SKIP IT when:

- You shot in Rec.709 (standard video mode)
- Your footage already looks "normal" and responds well to grading
- You're doing a quick grade on social media content shot on Auto

Step-by-Step Setup

1. **Select the N1 node** (click on it in the node graph)
2. **Open Effects Library** (top left of screen, or Workspace → Effects Library)
3. **Navigate:** OpenFX → ResolveFX Color → Color Space Transform
4. **Drag the effect onto your N1 node**
5. **In the Inspector panel (top right), set:**
    - Input Color Space: Your camera's color space
    - Input Gamma: Your camera's LOG curve
    - Output Color Space: DaVinci Wide Gamut
    - Output Gamma: DaVinci Intermediate

Common Camera Settings

Camera	Input Color Space	Input Gamma
Sony FX3/FX6/A7S III	S-Gamut3.Cine	S-Log3
Sony older (A7S II)	S-Gamut3	S-Log2
Blackmagic (Gen 5)	Blackmagic Design	Blackmagic Film Gen 5

Camera	Input Color Space	Input Gamma
Blackmagic (older)	Blackmagic Design	Blackmagic Film Gen 4
Canon C70/C300 III	Cinema Gamut	Canon Log 3
Canon older	Cinema Gamut	Canon Log 2
Panasonic S1H/GH6	V-Gamut	V-Log
ARRI (modern)	ARRI Wide Gamut 4	LogC4
ARRI (older)	ARRI Wide Gamut 3	LogC3
RED (IPP2 - modern)	REDWideGamutRGB	Log3G10
RED (pre-IPP2 / legacy)	DRAGONcolor2	REDlogFilm
iPhone 15 Pro (Log)	Apple Log	Apple Log
DJI (D-Log M)	DJI D-Gamut	D-Log

How to Know If It's Working

Before transform: Image looks flat, gray, washed out (if LOG) **After transform:** Image has more contrast and color, but may look slightly weird or oversaturated

Don't worry if it looks a bit off - you're just converting the data. The next nodes will make it look good.

Common Mistakes

❌ **Adding contrast or saturation here** - This node is ONLY for color space conversion ❌ **Using the wrong input settings** - Check your camera's recording settings ❌ **Applying to Rec.709 footage** - This will make normal footage look terrible ❌ **Forgetting to set BOTH input AND output** - Leaving either on "Bypass" defeats the purpose

Extra Nodes to Consider (N1 variants)

N1b - Noise Reduction (before everything else): If your footage is noisy, add a Temporal Noise Reduction node BEFORE or immediately after the color space transform. Noise reduction works best on the original data before any contrast adjustments amplify the noise.

```
LOCATION: Effects Library → OpenFX → ResolveFX Refinement →
          Noise Reduction (drag onto a new node before N2)
SETTINGS: Start with Temporal NR, Threshold 10-25, keep Spatial low
```

N1c - Lens Correction: If you have significant lens distortion or chromatic aberration, fix it early.

```
LOCATION: Color page → OpenFX → ResolveFX Refinement →
          Chromatic Aberration or Dead Pixel Fixer
```

End State Checklist

✓ Footage is in a consistent color space
✓ No creative changes have been made
✓ Image responds predictably when you test the color wheels
✓ You haven't touched contrast or saturation

N2 — PRIMARY BALANCE

Purpose: Establish physical reality - make the image look like what your eyes would have seen if you were standing there. This is the most important node in your entire grade.

```
┌─────────────────────────────────────────────────────┐
│ TOOLS: Lift / Gamma / Gain / Offset + Temp / Tint   │
│ LOCATION: Color page → Bottom left panel → Color Wheels │
│          (icon looks like 4 circles)                │
│ Temp/Tint: Below the Offset wheel, or switch to Primaries Bars │
└─────────────────────────────────────────────────────┘
```

What This Node Actually Does

This is where you make the image look "correct" - proper exposure, accurate colors, realistic black levels. You're not making it pretty yet, you're making it true.

The goal is a "boring" image - one that looks completely natural, like you're looking through a clean window at the real world.

Understanding the Controls

OFFSET (the big wheel in the center or the fourth wheel):

- Moves the ENTIRE image brighter or darker
- Like adjusting the exposure dial on your camera
- Use this FIRST to get overall brightness roughly right

LIFT (shadows wheel - usually bottom left):

- Controls the darkest parts of your image
- Dragging UP = brighter shadows (more milky/washed out)
- Dragging DOWN = darker shadows (more crushed/contrasty)
- Pushing toward a color tints your shadows that color

GAMMA (midtones wheel - usually top):

- Controls the middle brightness values
- This is where skin tones live
- Dragging UP = brighter midtones
- Dragging DOWN = darker midtones

GAIN (highlights wheel - usually bottom right):

- Controls the brightest parts of your image
- Dragging UP = brighter highlights (can clip/blow out)
- Dragging DOWN = darker highlights

TEMP (color temperature):

- LEFT (negative) = Cooler/bluer
- RIGHT (positive) = Warmer/more orange
- Corrects for wrong white balance in camera

TINT:

- LEFT (negative) = More green
- RIGHT (positive) = More magenta
- Corrects for fluorescent lights or mixed lighting

Step-by-Step Workflow

Step 1: Fix White Balance First

Look at something in your image that should be neutral - a white shirt, gray concrete, a piece of paper.

If it looks:

- **Too orange/yellow:** Drag Temp LEFT (cooler)
- **Too blue:** Drag Temp RIGHT (warmer)
- **Too green:** Drag Tint RIGHT (add magenta)
- **Too magenta/pink:** Drag Tint LEFT (add green)

Eyedropper trick: In the Color Wheels panel, there's an eyedropper icon. Click it, then click on something that should be neutral gray in your image. Resolve will auto-correct the white balance. This is a starting point - adjust by eye afterward.

Step 2: Set Overall Exposure with Offset

Look at your Waveform scope (Cmd/Ctrl + Shift + W to open scopes).

The waveform shows brightness from bottom (0% = pure black) to top (100% = pure white).

- If the whole waveform is too low, drag Offset UP
- If the whole waveform is too high, drag Offset DOWN

Step 3: Set Your Black Point with Lift

Look at the bottom of your waveform. Your darkest shadows should:

- Touch or nearly touch 0% (true black)
- NOT be crushed flat against 0% (you'd lose shadow detail)
- NOT be floating above 0% (image will look washed out)

What "crushing" looks like: When you drag Lift too far down, watch the bottom of your waveform. If it flattens out and a bunch of different dark values all become the same pure black, you've "crushed" your shadows. You've lost detail. Back off.

Visual test: Look at the darkest areas of your image - a person's black hair, shadows under furniture, the inside of a doorway. Can you still see detail and texture? If it's all just a blob of black, you've crushed.

What "milky" looks like: If your Lift is too high, your blacks won't be black - they'll be dark gray. The image will look faded, low-contrast, like a print that's been left in the sun. The bottom of your waveform will float above 0%.

Step 4: Set Faces with Gamma

Skin tones should fall in specific brightness ranges:

Lighting Scenario	Skin Luma Target
Bright daylight	50-60%
Normal indoor	40-55%
Moody/night	30-45%
Dramatic shadows	20-35%

How to check: Hover over face in the viewer while looking at the waveform. Or use the Qualifier to isolate skin and see where it sits on the scope.

If faces are too dark, push Gamma UP. If too bright, pull Gamma DOWN.

Step 5: Control Highlights with Gain

Look at the top of your waveform. Your brightest areas should:

- Stay below 100% (not clipping)
- Typically peak around 85-95%

- Only hit 100% for actual light sources (sun, lamps)

What "clipping" looks like: When highlights hit 100% and flatten against the top of the waveform, you've lost detail. A white shirt becomes a blob with no fabric texture. The sky becomes a featureless white void. This is bad and often unfixable.

If your highlights are clipping, drag Gain DOWN until they drop below 100%.

Target Values Reference

Element	Waveform Target	What to Look For
Pure black in scene	0-2%	Deepest shadows just touching bottom
Dark shadows	5-15%	Detail still visible
Dark skin tones	35-45%	Rich, not muddy
Medium skin tones	45-55%	Natural, not blown
Light skin tones	55-65%	Not overexposed
White objects	80-90%	Texture visible
Specular highlights	90-100%	Only small points
Light sources	100%	OK to clip

Reading the Vectorscope for White Balance

The vectorscope shows color, not brightness. For a properly white-balanced image:

- The "blob" of your image data should be roughly centered
- If it's shifted toward orange/yellow, image is too warm
- If it's shifted toward blue/cyan, image is too cool
- If it's shifted toward green, you have green tint
- If it's shifted toward magenta, you have magenta tint

Common Mistakes

❌ **Making it "look good" instead of "look correct"** - Save pretty for later nodes ❌ **Crushing shadows for "cinematic" look** - You'll regret losing that detail ❌ **Over-correcting white balance** - A slight warm or cool cast can be intentional ❌ **Ignoring the scopes** - Your eyes adapt and lie; scopes tell the truth ❌ **Adjusting one shot in isolation** - Always compare to surrounding shots

How to Know If You're Done

Toggle the node bypass (Cmd/Ctrl + D). The image should go from "correctly exposed, proper colors" to "whatever the camera recorded."

The "after" should look:

- Properly exposed (not too dark, not too bright)
- Correct colors (whites are white, not orange or blue)
- Full tonal range (blacks are black, whites are white, with detail in between)
- Boring (no style, no mood, just reality)

Extra Nodes to Consider (N2 variants)

N2b - Secondary Exposure Fix (for specific areas): If part of your image needs different exposure than the rest (like a bright window or dark corner), create a parallel node with a Power Window to fix just that area.

```
HOW: Alt/Option + P for parallel node, add Window, adjust exposure
USE: Bright windows, dark backgrounds, blown-out skies
```

N2c - Skin Tone Isolation: If skin tones need separate treatment from the rest of the image:

```
HOW: Add serial node after N2, use Qualifier to select skin
LOCATION: Bottom center panel → Qualifier tab → HSL
TIP: Select skin, add slight warmth or adjust exposure independently
```

End State Checklist

✓ White/neutral objects look neutral (not orange, blue, green, or magenta)
✓ Darkest shadows are near 0% on waveform with detail retained
✓ Brightest highlights are below 100% on waveform (except light sources)
✓ Faces are at appropriate brightness for the scene
✓ Image looks "correct" even if it's boring
✓ No style or mood has been added yet

N3 — CONTRAST SHAPING

Purpose: Add depth and dimension to the flat, "correct" image from N2. Make it feel cinematic without making it look "graded."

```
┌──────────────────────────────────────────────────────────────┐
│ TOOL: Custom Curves (preferred)                              │
│ LOCATION: Color page → Bottom left panel → Curves            │
│         (icon looks like an S-curve)                         │
│ Make sure dropdown says "Custom" (not Hue vs Hue etc.)       │
│                                                              │
│ ALTERNATIVE: Contrast + Pivot sliders                        │
│ LOCATION: Color page → Primaries Bars → Contrast / Pivot     │
└──────────────────────────────────────────────────────────────┘
```

What This Node Actually Does

After N2, your image is technically correct but often feels flat - like a news broadcast rather than a movie. Contrast shaping adds the subtle tonal compression and expansion that makes images feel three-dimensional and cinematic.

The key insight: Film doesn't capture light linearly. It compresses highlights and lifts shadows slightly. Digital cameras capture light more literally, which can look harsh. This node adds back that organic feeling.

Understanding Curves

The curve graph shows input values (original brightness) on the X-axis and output values (resulting brightness) on the Y-axis.

- **Straight diagonal line:** No change (output = input)
- **Line above the diagonal:** That brightness range gets brighter
- **Line below the diagonal:** That brightness range gets darker
- **Steeper curve:** More contrast in that range
- **Flatter curve:** Less contrast in that range

The "S-Curve": An S-shaped curve increases contrast in midtones while compressing shadows and highlights. This is the classic cinematic contrast shape.

```
         ___-------- Highlights: rolled off (compressed)
        /
       /
      /   Midtones: steep (more contrast)
     |
     |
    /
___/         Shadows: lifted toe (compressed)
```

Step-by-Step Workflow

Step 1: Open Custom Curves

1. Click on your N3 node
2. Go to Curves panel (bottom left)
3. Make sure dropdown at top says "Custom" (not Hue vs Hue, etc.)
4. You should see a diagonal line in a square graph

Step 2: Lift the Toe (Shadow Compression)

Click on the curve near the bottom-left (around 10-15% on the X-axis).
Drag this point UP slightly.

What this does: The very darkest parts of your image won't go to pure black anymore. They'll land at a slightly lifted dark gray. This mimics how film handles shadows and prevents harsh, digital-looking blacks.

How much: Start subtle. Lift the shadows so they land around 3-8% instead of 0%. You can always add more.

Visual check: Look at the darkest areas of your image. Do they feel rich and detailed, or flat and digital? They should feel "dense" - dark but with depth, not like a black hole.

Step 3: Roll Off the Highlights (Highlight Compression)

Click on the curve near the top-right (around 85-90% on the X-axis).
Drag this point DOWN slightly.

What this does: The brightest parts of your image won't go to pure white anymore. They'll compress into a softer, creamier highlight. This prevents harsh clipping and mimics how film handles bright areas.

How much: Roll off gently so highlights peak around 95-98% instead of 100%. You want them soft, not dull.

Visual check: Look at bright areas - windows, sky, white clothing. Do they feel soft and natural, or harsh and blown? They should feel luminous but not glaring.

Step 4: Shape the Midtones (Optional S-Curve)

If you want more punch, add a slight S-shape through the midtones:

1. Add a point around 30-35% (lower midtones) - drag slightly DOWN
2. Add a point around 65-70% (upper midtones) - drag slightly UP

This increases contrast in the midtones where faces live, making the image pop.

How much: Very subtle. If you can obviously see the effect, you've gone too far.

Understanding "Contrast" and "Pivot" Sliders

If curves feel too complex, you can use the simpler Contrast and Pivot sliders:

Contrast slider:

- Increases = more difference between darks and lights
- Decreases = flatter, less difference

Pivot slider:

- Sets the "center point" around which contrast is applied
- Lower pivot = contrast affects shadows more
- Higher pivot = contrast affects highlights more

For a cinematic look, try: Contrast +5 to +15, Pivot around 0.35-0.45

What Is "Crushing" and How to Avoid It

Crushing = losing shadow detail by pushing darks to pure black

When you add too much contrast or push your curve too far, the bottom of the waveform flattens. Multiple different dark gray values all become the same pure black. You've lost information permanently.

How to spot crushing:

- Waveform: Bottom flattens into a thick line at 0%
- Image: Shadow areas become featureless black blobs
- Details like hair texture, fabric in shadows, background details disappear

How to avoid it:

- Always lift the toe slightly (don't let curve touch bottom-left corner)
- Watch the waveform as you adjust
- Look at actual shadow detail in the image, not just overall contrast

Fix if you've crushed: Back off your adjustment. Pull that lower curve point back up until shadow detail returns.

What Is "Banding" and How to Avoid It

Banding = visible steps in smooth gradients instead of smooth transitions

Common in skies, out-of-focus backgrounds, and any smooth gradient area. Instead of a smooth blue-to-lighter-blue sky, you see distinct bands like steps.

How to spot banding:

- Look at skies and smooth gradients
- Visible "steps" or "rings" in areas that should be smooth
- Most obvious in 8-bit footage or heavily graded images

How to avoid it:

- Use gentle curves, not extreme adjustments
- Don't add excessive contrast to already-compressed footage

- Work in a higher bit-depth project if possible (10-bit or higher)

Fix if you have banding: Reduce the intensity of your contrast curve. Add a touch of grain/noise in N7 (grain hides banding).

Common Mistakes

❌ **Too much S-curve** - Image looks crunchy and over-processed ❌ **Crushing shadows for "mood"** - You're just losing information ❌ **Clipping highlights** - Watch the top of your waveform ❌ **Making faces too contrasty** - Faces should be lifted gently, not punchy ❌ **Adjusting contrast without looking at scopes** - Your eyes lie after a few minutes

How to Know If You're Done

Toggle the node bypass (Cmd/Ctrl + D).

The difference should be:

- Subtle but noticeable
- Image has more "depth" and dimension
- Shadows feel rich, not crushed
- Highlights feel soft, not blown
- It should NOT look "filtered" or "graded"

The "90% rule": If someone who doesn't do color grading notices the change, you've probably gone too far.

Extra Nodes to Consider (N3 variants)

N3b - RGB Contrast Separation: Instead of one contrast curve for all channels, you can use separate curves for Red, Green, and Blue. This creates subtle color separation between shadows and highlights.

```
HOW: In Curves panel, click "Y" and change to "R", "G", or "B"
TIP: Slightly different curves per channel = organic color separation
```

N3c - Zone-Based Contrast: For more control, use separate nodes for shadow contrast and highlight contrast.

```
HOW: Create two serial nodes
NODE 1: Only affect shadows (use Luminance Qualifier or low-end curve)
NODE 2: Only affect highlights (use high-end curve)
```

N3d - Local Contrast (use sparingly): If you want more "pop" without affecting overall contrast:

```
TOOL: Midtone Detail (positive values)
LOCATION: Blur panel → Mid Detail slider
WARNING: Easy to overdo. Keep it subtle (+5 to +15)
```

End State Checklist

✓ Image has depth and dimension (not flat)
✓ Shadows are rich and detailed (not crushed to black)
✓ Highlights are soft and natural (not clipped to white)
✓ No visible banding in gradients
✓ Faces look natural (not over-contrasty)
✓ Change is subtle - wouldn't look "filtered" to a normal viewer

---

N4 — HIGHLIGHT SAFETY

Purpose: Protect highlights from getting weird colors when you add saturation later. This is preventive medicine for common digital artifacts.

```
┌─────────────────────────────────────────────────────┐
│ TOOL: Luma vs Sat curve                              │
│ LOCATION: Color page → Bottom left panel → Curves →  │
│         Click dropdown at top → Select "Lum vs Sat"  │
│                                                      │
│ The X-axis is luma (brightness), Y-axis is saturation │
│ Pull down the RIGHT side (highlights) to desaturate brights │
└─────────────────────────────────────────────────────┘
```

What This Node Actually Does

When you add saturation to an image (in N5), the brightest areas can get weird. Skies turn electric blue. White shirts get yellow or magenta casts. Skin highlights go orange.

This node preemptively reduces saturation in the highlights so that when you boost overall saturation later, the bright areas stay clean and neutral.

Think of it like sunscreen: You apply it before you go outside, not after you're burned.

The Problem This Solves

Why highlights get weird with saturation:

Digital sensors capture color differently than film. When a pixel gets very bright, it can lose accurate color information. The "color" you see in those highlights is often just noise or sensor artifacts, not real color data.

When you boost saturation, you're amplifying everything - including those artifacts. A slightly yellow-ish highlight becomes VERY yellow. A slightly blue sky goes nuclear.

Common highlight problems:

- **Sky turning electric/neon blue** - Most common, looks terrible
- **White clothing getting yellow or magenta tint**

- **Skin highlights going orange** - Makes people look fake
- **Specular reflections picking up weird colors**
- **Light sources (windows, practicals) getting color casts**

Understanding the Luma vs Sat Curve

This curve lets you control saturation based on brightness:

- **X-axis (left to right):** Brightness from 0% (black) to 100% (white)
- **Y-axis (up and down):** Saturation adjustment
- **Center line:** No change
- **Above center:** More saturation
- **Below center:** Less saturation

Default state: Flat horizontal line in the middle = no change to any brightness range

Step-by-Step Workflow

Step 1: Open Luma vs Sat

1. Click on your N4 node
2. Go to Curves panel (bottom left)
3. Click the dropdown at the top (probably says "Custom")
4. Select "Lum vs Sat"

Step 2: Anchor Your Midtones

Before touching anything, add anchor points to protect the areas you DON'T want to change:

1. Click on the curve around 40-50% (midtones) to add a point
2. Click around 60-70% (upper midtones) to add another point
3. Don't move these points - they're anchors

Why anchor: Without anchors, when you pull down the highlights, the change affects more of the image than you want. Anchors keep the midtones (where skin lives) untouched.

Step 3: Pull Down the Highlights

1. Click on the curve at the far right (around 90-95%)
2. Drag this point DOWN
3. The curve should now dip down on the right side only

How much: Start subtle. Pull down until the very brightest areas lose maybe 20-30% of their saturation. You can always do more if needed.

Step 4: Optional - Protect the Extreme Shadows

Sometimes very dark shadows also get noisy/weird saturation. You can pull down the left side slightly too:

1. Click on the curve at the far left (around 5-10%)
2. Drag this point DOWN slightly
3. Add an anchor around 20% to protect lower midtones

How to Check If It's Working

The Before/After Test:

1. Go to N5 and temporarily boost saturation to an extreme amount (like +30)
2. Look at your highlights - are they going crazy?
3. Now toggle N4 on and off (Cmd/Ctrl + D)
4. With N4 ON: Highlights should stay relatively neutral
5. With N4 OFF: Highlights will get colorful/weird
6. Reset N5 saturation when done testing

What to Look At:

Element	Without N4 (bad)	With N4 (good)
Sky	Electric/neon blue	Clean, natural blue
White clothing	Yellow or magenta tint	Clean white with texture
Skin highlights	Orange, unnatural	Soft, natural glow
Windows/lights	Color cast	Clean, white-ish

Common Mistakes

❌ **Pulling down too much** - Highlights become gray and lifeless ❌ **Forgetting to anchor midtones** - Affects skin and other important areas ❌ **Skipping this node** - "I'll fix it if there's a problem" - by then it's harder ❌ **Not testing with saturation boost** - You won't see the effect until you saturate

How Extreme Is Too Extreme?

If your highlights look:

- **Gray and dead:** You've pulled down too much. Back off.
- **Clean but slightly less colorful than the rest of the image:** Perfect.
- **Still getting neon/weird colors when you add saturation:** Pull down more.

The goal is "invisible protection" - you shouldn't be able to see what this node does until you stress-test it with saturation.

Extra Nodes to Consider (N4 variants)

N4b - Highlight Color Correction: If highlights have a specific color cast (like yellow), you might want to correct the color, not just desaturate.

```
HOW: Add serial node, use Qualifier or Curves to select highlights
THEN: Adjust Temp/Tint or color wheels to neutralize the cast
```

N4c - Highlight Recovery (for clipped footage): If your highlights are already blown out from shooting:

```
TOOL: Highlight (HDR) recovery in Camera RAW panel (for RAW footage)
OR: Use the Gain or Highlight slider to pull down brightness
LIMITATION: Can't recover what wasn't captured
```

N4d - Sky Replacement/Fix Node: If sky is particularly problematic, create a dedicated sky node:

```
HOW: Add node, use Qualifier to select sky (HSL: select blues/cyans)
THEN: Adjust saturation, hue, and luminance just for sky
TIP: Feather the selection to avoid harsh edges
```

End State Checklist

✓ Midtones are anchored and unchanged
✓ Highlights have reduced saturation (you may not be able to see it yet)
✓ When you boost saturation in N5, highlights stay clean
✓ Sky doesn't go nuclear blue when saturated
✓ White objects stay white, not tinted
✓ The node appears to "do nothing" under normal viewing

N5 — SATURATION & COLOR DENSITY

Purpose: Bring life and energy back into the image after all the technical corrections. Make the world feel vibrant and alive without looking "processed."

```
┌─────────────────────────────────────────────┐
│ TOOLS: Saturation slider + Color Boost slider │
│ LOCATION: Color page → Bottom left panel → Primaries Bars │
│         (click the sliders icon, not the wheels)       │
│                                             │
│ Saturation: Main "Sat" slider at the bottom of the panel │
│ Color Boost: Right below/beside Saturation slider     │
└─────────────────────────────────────────────┘
```

What This Node Actually Does

After color space conversion and balancing, images often feel a bit flat or desaturated. This node adds back color richness - the feeling of lush greens, deep blues, warm skin tones.

The goal: The world should feel alive and present, like you could reach into the screen and touch things. But it should NOT feel like an Instagram filter.

Understanding the Two Tools

SATURATION (main slider):

- Increases/decreases ALL colors equally
- +10 = everything gets 10% more saturated
- Affects already-saturated colors as much as desaturated ones
- Can push skin and vibrant colors too far quickly

COLOR BOOST (secondary slider):

- Smart saturation that protects already-saturated areas
- Boosts undersaturated colors more than saturated ones
- Gentler on skin tones
- Good for bringing up weak colors without over-cooking strong ones

When to use which:

Situation	Use	Why
Image is evenly desaturated	Saturation	Everything needs equal boost
Some colors weak, some OK	Color Boost	Brings up weak without over-doing strong
Skin tones present	Color Boost first	More forgiving on skin
Want vibrant, punchy look	Saturation	More aggressive effect
Subtle, natural look	Color Boost	More controlled

Step-by-Step Workflow

Step 1: Open Primaries Bars

1. Click on your N5 node
2. Go to the bottom-left panel
3. Click the "bars" icon (looks like horizontal sliders) instead of the wheels

Step 2: Start with Color Boost

1. Find the Color Boost slider (usually labeled "Color Boost" or near Saturation)
2. Start at 0
3. Slowly push up to +10 or +15
4. Watch the image, especially skin and already-colorful areas

Step 3: Add Saturation if Needed

1. If Color Boost isn't enough, add some Saturation
2. Start at 50 (default)
3. Push up to 52-58
4. Watch for skin going orange and shadows getting noisy

Step 4: Check Your Danger Zones

Look at these areas specifically:

Danger Zone	What Can Go Wrong	What to Watch For
Skin tones	Goes orange/red, looks fake	Should look human, not tanned
Shadows	Chroma noise appears	Colored speckles in dark areas
Sky	Goes electric blue	Should be natural blue
Foliage	Goes neon green	Should be organic, not radioactive
Reds	Blows out first	Red clothing, lips, etc.

Understanding Chroma Noise

What is chroma noise? Random colored speckles (usually red/blue/magenta/green) that appear in dark areas of the image. Looks like confetti in your shadows.

Why saturation causes it: Shadow areas have weak color information from the sensor. When you boost saturation, you amplify this weak signal, making the noise visible.

How to spot it:

1. Look at the darkest areas of your image at 100% zoom
2. Look for random colored pixels that shouldn't be there
3. Most visible in: dark clothing, night scenes, underexposed areas

How to avoid it:

- Don't over-saturate
- Use Color Boost instead of raw Saturation
- If you see it, back off Saturation or add noise reduction
- You can also use Lum vs Sat to reduce saturation in shadows (like you did for highlights in N4)

Typical Settings

Look	Saturation	Color Boost	Notes
Natural/documentary	50	5-10	Minimal boost
Standard narrative	52-55	10-15	Healthy but not pushed
Vibrant commercial	55-60	15-20	Punchy but not garish
Music video/stylized	60-70	10-15	Intentionally saturated
Desaturated/gritty	40-48	0	Intentionally muted

Reading the Vectorscope

The vectorscope shows color saturation and hue:

- **Distance from center = saturation** (more saturated = further out)
- **Angle around circle = hue** (colors are labeled around the edge)
- **Skin tone line** = the line pointing toward "red/yellow" area (about 11 o'clock)

As you add saturation, the "blob" of your image data should expand outward from the center.

Watch for:

- Any color hitting the edge of the scope (over-saturated)
- Skin tones drifting off the skin tone line (going too orange or too pink)
- Overall shape becoming lopsided (one color much more saturated than others)

Common Mistakes

❌ **Eyeballing without scopes** - Your eyes adapt and you'll over-saturate ❌ **Boosting saturation to compensate for flat contrast** - Fix contrast in N3 instead ❌ **Ignoring shadow noise** - Zoom in and look at the dark areas ❌ **Making skin look "healthy" by adding saturation** - Skin should be human, not tanned ❌ **Trying to fix hue problems with saturation** - Hue fixes belong in N6

How to Know If You're Done

The "Is this a photo or real life?" test: Look away from the screen for 30 seconds (rest your eyes), then look back. Does the image look:

- Natural and believable? ✓
- Like it has a filter? ✗ (back off saturation)
- Still a bit flat? (add a bit more, carefully)

The bypass test (Cmd/Ctrl + D): The difference should be:

- Noticeable but not dramatic
- Makes the image feel more alive
- Doesn't make you go "whoa, that's colorful"

Extra Nodes to Consider (N5 variants)

N5b - Shadow Saturation Control: If shadows are getting noisy but you want more saturation in midtones:

```
HOW: Add another Lum vs Sat curve node
CURVE: Pull down the LEFT side (shadows) to desaturate darks
KEEP: Midtones and highlights at normal
```

N5c - Saturation by Color (Hue vs Sat): If you want different saturation amounts for different colors:

```
TOOL: Hue vs Sat curve
LOCATION: Curves panel → dropdown → Hue vs Sat
HOW: Boost saturation for weak colors, reduce for strong colors
```

N5d - Vibrance vs Saturation: Resolve doesn't have a "Vibrance" slider like Lightroom, but Color Boost is similar. If you want more control:

```
HOW: Use a Parallel Node structure
NODE A: High saturation on midtones only (use qualifier)
NODE B: Low saturation on already-saturated areas
BLEND: Mix to taste
```

End State Checklist

✓ Image feels alive and present (not flat)
✓ Skin tones look human (not orange or fake-tanned)
✓ No visible chroma noise in shadows
✓ Sky and foliage look natural (not neon)
✓ No colors hitting the edge of the vectorscope
✓ Looks like reality, not Instagram

N6 — HUE SCULPT (SELECTIVE COLOR CONTROL)

Purpose: Fine-tune individual colors to make the world feel lush, intentional, and subtly controlled. This is where you decide which colors matter and shape them precisely.

```
┌─────────────────────────────────────────────────────────────┐
│ TOOLS: Hue vs Hue / Hue vs Sat / Hue vs Lum curves          │
│ LOCATION: Color page → Bottom left panel → Curves →         │
│           Click dropdown at top → Select curve type         │
│                                                             │
│ Hue vs Hue: Shifts one hue toward another (e.g., cyan → blue) │
│ Hue vs Sat: Boosts/reduces saturation of specific hues      │
│ Hue vs Lum: Brightens/darkens specific hues                 │
│                                                             │
│ HOW TO USE: Click on the curve to add a point, drag to adjust │
│ TIP: Ctrl/Cmd + click on the image to auto-select that hue  │
└─────────────────────────────────────────────────────────────┘
```

What This Node Actually Does

N5 was about "how much color" overall. N6 is about "which specific colors" and where they point.

Common uses:

- Making sky a cleaner, more pleasant blue
- Making foliage a healthier green
- Removing ugly color pollution (like magenta in shadows)
- Unifying the color palette

The goal: A world that feels cohesive and intentional, where no color distracts or feels "off."

Understanding the Three Curve Types

HUE vs HUE (Steering Colors)

- X-axis: Input hue (the color wheel, left to right)

- Y-axis: Output hue offset (how much to shift)
- Use: Changing one hue toward another (cyan → blue, yellow-green → green)

HUE vs SAT (Selective Saturation)

- X-axis: Hue (the color wheel)
- Y-axis: Saturation adjustment for that hue
- Use: Boosting or reducing saturation of specific colors only

HUE vs LUM (Selective Brightness)

- X-axis: Hue (the color wheel)
- Y-axis: Luminance adjustment for that hue
- Use: Making specific colors brighter or darker (use sparingly)

The Color Wheel Layout

The hue axis follows the color wheel:

```
Left side → Middle → Right side
Red → Orange → Yellow → Green → Cyan → Blue → Magenta → Red

Approximate positions:
0° / 360°: Red
30°: Orange
60°: Yellow
120°: Green
180°: Cyan
240°: Blue
300°: Magenta
```

Step-by-Step Workflow

Step 1: Identify Problem Colors

Before touching anything, look at your image and ask:

- Is any color ugly or distracting?
- Is the sky a weird cyan instead of blue?
- Is foliage too yellow or too blue-green?
- Are there magenta or green casts in shadows?

Step 2: Use Hue vs Hue for Color Steering

Most common fix - Cleaning up sky:

Skies often record as cyan (blue-green) instead of pure blue. To fix:

1. Select "Hue vs Hue" from dropdown
2. Ctrl/Cmd + click on the sky in your image - this auto-selects that hue
3. Three points appear on the curve
4. Drag the CENTER point DOWN to push cyan toward blue
5. The outer two points are "boundaries" - they limit the effect

How much to move: Very small amounts. Moving 5-10 degrees is a lot. If you move 20°, you've gone way too far.

Visual check: Does the sky look like a sky you'd see in real life? Not gray, not electric, just... sky blue?

Step 3: Use Hue vs Sat for Selective Energy

After steering colors, you might want to boost or reduce specific ones:

Boosting weak colors:

1. Select "Hue vs Sat"
2. Ctrl/Cmd + click on a color you want to boost
3. Drag the center point UP to add saturation to just that hue

Reducing distracting colors:

1. Ctrl/Cmd + click on the distracting color
2. Drag the center point DOWN to reduce its saturation

Common Hue vs Sat moves:

Color	Direction	Why
Sky blues	Slight UP	Make sky feel present
Foliage greens	Slight UP	Lush, healthy vegetation
Skin	DO NOT TOUCH	Seriously, don't
Magentas	DOWN	Remove pollution/artifacting
Cyans in shadows	DOWN	Reduce color cast

Step 4: Protect Skin Tones (CRITICAL)

Skin tones live in a specific hue range (roughly orange/red, around 15-40° on the hue circle).

NEVER adjust this region in Hue vs Hue. Even small shifts make skin look sick or fake.

How to protect skin:

- Look at where your adjustment points are
- Make sure your curve is FLAT through the skin tone region
- Add anchor points on either side of the skin region if needed
- Test by looking at faces - any shift should be invisible

Skin protection visualization:

```
    DO NOT TOUCH THIS REGION
         ↓
  ┌─────────────────────────┐
──│          │──
  └─────────────────────────┘
~15°      ~40°
(Red)     (Orange)
```

Common Color Problems and Fixes

Problem	Tool	Fix
Sky is too cyan	Hue vs Hue	Pull cyan region toward blue (down)
Sky is dull/gray	Hue vs Sat	Boost saturation in blue/cyan region
Grass is too yellow	Hue vs Hue	Pull yellow-green toward green
Grass is neon green	Hue vs Sat	Reduce saturation in green region
Skin is too orange	N2 (white balance)	Fix in primary balance, not here
Magenta in shadows	Hue vs Sat	Reduce saturation in magenta region
Everything feels same-y	Hue vs Hue	Create subtle separation between colors

The "It Looks Graded" Warning

If you can tell colors have been pushed, you've gone too far.

This node should be invisible to normal viewers. The sky should just look like a nice sky, not a "color graded sky."

Signs you've overdone it:

- Colors look "selected" or isolated
- The image feels artificial or "digital"

- Someone watching would say "that sky is really blue"
- Faces look strange (even slightly)

Fix: Reduce all your adjustments by 50% and see if it still works.

Common Mistakes

❌ **Moving hues too far** - 5-10° is usually plenty, 20°+ is almost always wrong ❌ **Touching skin tones** - This is not where skin issues get fixed ❌ **Making every color "better"** - The goal is cohesion, not perfection ❌ **Using Hue vs Lum aggressively** - Changes brightness weirdly, use sparingly ❌ **Forgetting to create boundaries** - Without outer anchor points, changes bleed into other colors

How to Know If You're Done

The "turn it off" test: Toggle N6 bypass. The difference should be:

- Barely noticeable to a casual viewer
- You notice the sky is slightly better/cleaner
- You notice foliage feels slightly more natural
- Skin looks exactly the same

The "someone else" test: Show someone the image without telling them you graded it. Ask "does anything look weird or off?" If they say no, you're good.

Extra Nodes to Consider (N6 variants)

N6b - Qualifier-Based Color Isolation: For more precise control than curves allow:

```
TOOL: Qualifier (HSL)
LOCATION: Bottom center panel → Qualifier tab
HOW: Use eyedropper to select specific color, adjust only that selection
USE: When curves affect too broad a range
```

N6c - Color Warper: Alternative to Hue vs Hue curves with a visual grid interface:

```
TOOL: Color Warper
LOCATION: Color page → Bottom left panel → Color Warper tab (grid icon)
HOW: Drag color points on the grid to shift them
USE: When you want to visualize color relationships spatially
```

N6d - RGB Mixer (for extreme color control): For technical color separation:

```
TOOL: RGB Mixer
LOCATION: Color page → Primaries panel → RGB Mixer tab
USE: Advanced color channel mathematics - use only if you know what you're doing
```

N6e - Skin Isolation Node: If skin needs its own treatment:

```
HOW: Add parallel node, use Qualifier to select skin only
THEN: Adjust hue, saturation, luminance for skin independently
TIP: Use very soft selection and check edges carefully
```

End State Checklist

✓ Sky looks like a pleasant, natural sky
✓ Foliage looks healthy, not neon
✓ No distracting color casts in shadows
✓ Skin tones are UNCHANGED from N5
✓ A normal viewer wouldn't notice colors have been adjusted
✓ The world feels cohesive and intentional

N7 — TEXTURE CONTROL

Purpose: Control the perceived "feel" of the image - smooth vs. detailed, digital vs. filmic. This is subtle but makes a significant difference in how professional the image looks.

```
┌─────────────────────────────────────────────────────────────┐
│ TOOL: Midtone Detail (aka Mid/Detail or MD)                 │
│                                                             │
│ LOCATION 1: Color page → Primaries Wheels/Bars →            │
│           Look for "Mid/Detail" slider in the adjustment    │
│           controls (same row as Contrast, Pivot, etc.)      │
│                                                             │
│ LOCATION 2: Color page → Log Wheels →                       │
│           Bottom bar → "Mid Detail" or "MD" slider          │
│           (next to Color Boost, Shad, Hi/Light)             │
│                                                             │
│ NOTE: Both locations control the SAME parameter - use whichever │
│      panel you're already working in. They are not independent. │
│                                                             │
│ SHARPEN (different tool, use sparingly):                    │
│ LOCATION: Color page → Blur panel (water drop icon) →       │
│          Sharpen section → H/V sliders                      │
└─────────────────────────────────────────────────────────────┘
```

What This Node Actually Does

Modern digital cameras and phones apply aggressive processing that can make images look "digital" - overly sharp, crunchy, with weird halos around edges. This node fixes that.

Midtone Detail adjusts local contrast in the midtones:

- **Negative values:** Soften the image, reduce harsh digital texture
- **Positive values:** Add perceived detail and "pop"

Think of it like this: You're not blurring or sharpening the whole image - you're adjusting how "crunchy" vs. "smooth" the texture feels.

Understanding Midtone Detail

What "midtone detail" actually controls: It's a local contrast adjustment that affects texture perception. When you reduce it, fine details become smoother. When you increase it, fine details pop more.

Negative Midtone Detail (-5 to -20):

- Smooths skin texture (without blurring features)
- Removes harsh digital processing look
- Makes the image feel more "filmic" or "organic"
- Good for: beauty work, skin, soft natural scenes

Positive Midtone Detail (+5 to +20):

- Enhances perceived detail and texture
- Makes surfaces pop (brick, metal, fabric)
- Adds "crunch" and definition
- Good for: landscapes, architecture, product shots, gritty scenes

How to Find Midtone Detail

This confuses people because it's not where you'd expect:

1. Go to Color page
2. Look at the bottom-left panel (where Color Wheels, Curves, etc. are)
3. Find the icon that looks like a water drop (Blur panel)
4. Click it
5. Look for "Mid Detail" slider (might be abbreviated "MD")

If you can't find it:

- Try different panel layouts (drop-down arrow at top of panel)
- It's in the same area as Blur, Sharpen, and Mist
- It's NOT in Effects Library - it's built into the Color page

Step-by-Step Workflow

Step 1: Assess Your Image

Look at your image at 100% zoom and ask:

- Does skin look harsh or pore-y? → Reduce MD
- Do textures look flat or mushy? → Increase MD
- Does the image feel "digital"? → Probably reduce MD
- Is it from a phone that over-processed? → Reduce MD

Step 2: Make Your Adjustment

For most footage:

Source	Starting Point	Why
Cinema camera (ARRI, RED)	0 to +5	Already looks good, maybe slight boost
Sony/Blackmagic	0 to +5	Generally clean, might want subtle pop
Panasonic	-5 to 0	Can be slightly harsh
Canon DSLR	-10 to 0	Often over-sharpened in camera
iPhone/phone	-15 to -5	Heavy processing to undo
GoPro	-10 to -5	Aggressive sharpening
Drone (DJI)	-5 to +5	Varies by model

Step 3: Check Specific Areas

Zoom to 100% and check:

Area	What to Look For	Fix
Skin	Pores visible but not emphasized	Reduce MD if pores are harsh
Hair	Natural texture, no halo	Reduce if haloing appears
Fabric	Texture visible, not crunchy	Adjust to taste
Eyes	Sharp but not weird	Should be naturally detailed
Background	Smooth gradients	Check for artifacts

What Are "Halos" and How to Avoid Them

Halos = bright or dark edges around objects, especially against contrasting backgrounds

You'll see them as:

- Light line around a dark object against a bright background
- Dark line around a bright object against a dark background
- "Glow" around hair or tree branches against sky

What causes halos:

- Over-sharpening
- Too much positive Midtone Detail
- Aggressive in-camera processing

How to fix:

- Reduce Midtone Detail
- Don't use Sharpen (or use very little)
- Accept that some detail is less important than avoiding artifacts

Understanding the Blur Panel Controls

The Blur panel has several controls:

Radius (Blur):

- Actual blur amount
- Usually leave at 0 unless you want intentional softening
- Use with windows for selective blur

H/V Sharpen:

- Horizontal and Vertical sharpening
- Very aggressive - use sparingly (2-8 max)
- Creates halos easily
- Only use if image is genuinely soft

Mist:

- Adds halation/glow to highlights
- Creates a dreamy, filmic look
- Usually 0-10% if used at all

Mid Detail:

- What we've been discussing
- -50 to +50 range

- Most useful control for texture

Scaling:

- Controls the "size" of the detail adjustment
- Usually leave at default
- Lower = finer detail, Higher = broader detail

Common Mistakes

❌ **Using Sharpen instead of Mid Detail** - Sharpen creates halos, Mid Detail is more organic ❌ **Going too extreme** - ±20 is usually maximum, ±10 is typical ❌ **Not checking at 100% zoom** - You can't see texture issues zoomed out ❌ **Applying same value to all shots** - Different sources need different treatment ❌ **Using Clarity** - If your version of Resolve has it, avoid - creates unnatural local contrast

How to Know If You're Done

The zoom test:

1. View at 100% zoom
2. Look at skin - does it look like real skin? Not waxy, not harsh?
3. Look at fabric - can you see the texture without it being distracting?
4. Look at hair - natural edges, no halos?
5. Look at sky/gradients - smooth, no artifacts?

The bypass test (Cmd/Ctrl + D): The change should be:

- Subtle but real
- Makes the image feel more "organic" or "film-like" (if reducing)
- Makes texture more present (if increasing)
- Not dramatically different

Extra Nodes to Consider (N7 variants)

N7b - Skin Softening (selective): If skin needs more softening than the rest of the image:

```
HOW: Add serial node, use Qualifier to select skin tones
THEN: Add blur (2-4 radius) or more negative Mid Detail
TIP: Keep it subtle - waxy skin looks worse than textured skin
```

N7c - Detail Enhancement (selective): If you want to boost detail in specific areas only:

```
HOW: Add serial node, use Power Window to select area
THEN: Boost Mid Detail or add slight Sharpen (2-4)
USE: Eyes, product details, architectural elements
```

N7d - Grain Addition: Film grain can hide digital artifacts and add organic feel:

```
TOOL: Film Grain (OpenFX)
LOCATION: Effects Library → OpenFX → ResolveFX Texture → Film Grain
SETTINGS: Start subtle - Grain Amount 5-15, Size 1-2
TIP: Grain also helps hide banding in gradients
```

N7e - Noise Reduction (if not done in N1): If you still have noise issues:

```
TOOL: Noise Reduction
LOCATION: Effects Library → OpenFX → ResolveFX Refinement → Noise Reduction
SETTINGS: Temporal NR first (Threshold 10-25), Spatial NR sparingly
WARNING: Too much NR = waxy, artificial look
```

End State Checklist

✓ Skin looks natural (not waxy, not harsh)
✓ No halos around edges (especially hair against sky)
✓ Textures feel organic, not digital
✓ Image has appropriate "feel" for its content
✓ Change is subtle - wouldn't be noticed by casual viewer

N8 — VIGNETTE / EYE GUIDANCE (OPTIONAL)

Purpose: Subtly guide the viewer's eye toward the subject without them noticing. This is invisible compositional control.

```
┌─────────────────────────────────────────────────────────────┐
│ TOOL: Power Window (Circle)                                 │
│ LOCATION: Color page → Bottom CENTER panel → Window tab      │
│           → Click the Circle icon to add a circular window  │
│                                                             │
│ THEN: Use the Lift/Gamma/Gain wheels to darken the area     │
│      OR use the Master Wheel (large wheel) in the center    │
│                                                             │
│ SOFTNESS: In the Window panel, use "Soft" sliders (1-4)     │
│          to feather the edge                                │
│                                                             │
│ INVERT: Click the "Invert" button to affect OUTSIDE the window │
└─────────────────────────────────────────────────────────────┘
```

What This Node Actually Does

A vignette subtly darkens the edges of the frame, drawing attention toward the center (where your subject usually is). When done right, viewers never notice it - they just naturally look where you want them to look.

The key word is "invisible." If anyone notices the vignette, you've done it wrong.

When to Use This Node

USE IT when:

- Your composition has a clear subject you want to emphasize
- The edges of frame are distracting
- You want a slightly more "finished" feel
- The subject doesn't move significantly

SKIP IT when:

- Wide shots with important edge content
- Fast-moving subjects
- Documentary/natural style
- The shot already has natural vignetting from the lens

Step-by-Step Workflow

Step 1: Add a Circle Window

1. Click on your N8 node
2. Go to the bottom-CENTER panel (not bottom-left)
3. Click "Window" tab
4. Click the Circle icon

A circle overlay appears on your image.

Step 2: Position and Size the Window

1. Drag the center point to position (usually near subject's face)
2. Drag the corner handles to resize
3. Make it large enough to cover the subject and important surroundings
4. Usually covers 60-80% of the frame

Step 3: Invert the Selection

By default, the window affects INSIDE the circle. You want to darken OUTSIDE.

1. Look for "Invert" button in the Window panel
2. Click it - now your adjustments affect only the edges of frame

Step 4: Add Softness (CRITICAL)

This is the most important step. A hard-edged vignette looks terrible.

1. Find the "Soft" sliders (usually labeled 1-4)
2. Crank them UP - try 70-100 to start
3. You want the transition to be extremely gradual
4. No visible edge should exist

Step 5: Darken the Edges

Now use the color wheels to darken the selected (outside) area:

Method 1 - Master Wheel:

- Find the large wheel labeled "Master" or the center Offset-like control
- Drag it down slightly (darker)
- Amount: 0.15-0.25 stops (very subtle)

Method 2 - Gamma:

- Drag Gamma wheel down slightly
- Affects midtones more than shadows or highlights
- Often looks more natural than pure exposure reduction

Step 6: Check Your Work

1. Press Cmd/Ctrl + Shift + H to hide the window overlay
2. Look at the image - can you see the vignette?

3. If yes, either add more softness or reduce the darkening
4. Toggle node bypass - the change should be barely perceptible

How Dark Is Too Dark?

Good vignette:

- You can't see where it starts or ends
- Edges feel naturally darker (like peripheral vision)
- Subject seems naturally more prominent
- Takes 5-10 seconds for someone to notice if you point it out

Bad vignette:

- Visible oval or circle shape
- Obviously darker corners
- Looks like an Instagram filter
- Anyone watching immediately thinks "vignette"

Typical amounts:

- Light vignette: 0.1-0.15 stops darker
- Standard vignette: 0.15-0.25 stops darker
- Heavy vignette (rarely appropriate): 0.25-0.4 stops darker

Using Tracking for Moving Subjects

If your subject moves significantly, the vignette should follow them:

Step 1: Enable tracking

1. In the Window panel, find "Tracker" section
2. Select tracking type (usually "Cloud Tracker" or point tracker)
3. Click "Track Forward" and/or "Track Reverse"

Step 2: Verify

- Play through the clip
- Make sure the center stays roughly on your subject
- Adjust tracking if it drifts

Using Shapes Other Than Circles

Power Windows come in several shapes:

Shape	Use Case
Circle/Oval	Most common, general vignette
Rectangle	Widescreen compositions
Polygon	Irregular shapes, avoiding specific elements
Gradient	Directional darkening (top-down, side)
Bezier	Complex, custom shapes

For vignettes, Circle/Oval is almost always correct.

Common Mistakes

❌ **Not enough softness** - Hard edges are immediately visible ❌ **Too much darkening** - Looks like a filter, not subtle guidance ❌ **Centered incorrectly** - Should focus on subject, not geometric center ❌ **Using on every shot** - Some shots don't need it ❌ **Visible corners** - If you can see the corners are dark, it's too much ❌ **Forgetting to invert** - You darken the subject instead of the edges

How to Know If You're Done

The "stare at it" test: Look at your image for 30 seconds. Does your eye naturally go to the subject? Do the edges feel like they're just... there, nothing special?

The "show someone" test: Show someone the image and ask where their eye goes. Don't mention vignette. If they look at the subject and don't mention the edges being dark, you've succeeded.

The bypass test (Cmd/Ctrl + D): The change should be:

- Almost imperceptible
- Makes the subject feel slightly more present
- Doesn't change the mood or feel dramatically

Extra Nodes to Consider (N8 variants)

N8b - Directional Falloff: Instead of circular vignette, darken from a direction (like simulating window light):

```
HOW: Use Gradient window instead of Circle
SHAPE: Linear gradient from one side
USE: Simulating directional light, drawing eye in specific direction
```

N8c - Subject Highlight: Instead of darkening edges, slightly brighten the subject:

```
HOW: Circle window (not inverted), slight Gamma lift
AMOUNT: Very subtle (+0.1 to +0.15 stops)
TIP: Even more invisible than darkening edges
```

N8d - Multiple Vignettes: For complex compositions:

```
HOW: Create multiple Circle windows in same node
OR: Use multiple N8 nodes with different windows
USE: Multiple subjects, avoiding bright elements at different edges
```

End State Checklist

✓ Window is inverted (affecting outside, not inside)
✓ Softness is very high (no visible edge)
✓ Darkening is subtle (0.15-0.25 stops)
✓ Center is positioned on subject (not frame center)
✓ Change is invisible unless pointed out
✓ Eye naturally goes to subject

N9 — CREATIVE LOOK

Purpose: Apply artistic intent - the "style" or "mood" that makes this project feel like itself. This is where you finally get to be creative.

```
┌──────────────────────────────────────────────────────────┐
│ MULTIPLE TOOLS - CHOOSE YOUR APPROACH:                     │
│                                                           │
│ LUT APPLICATION:                                          │
│ → Right-click on node → LUT → Browse for your .cube file  │
│ → Or: Color page → LUTs tab (top right) → drag onto node  │
│                                                           │
│ COLOR WHEELS (for manual look):                           │
│ → Color page → Bottom left → Color Wheels                 │
│ → Push Gain toward warm (orange), Lift toward cool (blue) │
│                                                           │
│ PER-CHANNEL CURVES (for advanced look):                   │
│ → Color page → Curves → Click "Custom" dropdown           │
│ → Select individual channels: Y (luma), R, G, or B        │
│ → Edit Blue channel: raise shadows, lower highlights = warm │
└──────────────────────────────────────────────────────────┘
```

What This Node Actually Does

Everything before N9 was about making the image technically correct and perceptually pleasing. N9 is about making it emotionally correct - adding the mood, style, and feeling that serves the story.

The key insight: Style only works when applied to a solid foundation. If your N1-N8 are good, almost any look will work. If they're bad, no look will save you.

Approaches to Creating a Look

Method 1: Using a LUT (Fastest)

A LUT (Look-Up Table) is a preset color transformation. Someone else already did the work.

To apply a LUT:

1. Right-click on N9 node
2. Select "LUT"
3. Browse to your .cube file
4. Click to apply

Important: LUTs are designed for specific input. Most expect:

- Rec.709 input (standard video)
- Properly balanced footage
- Correct exposure

If your footage isn't properly balanced, the LUT will look wrong.

Adjusting LUT intensity: LUTs at 100% are often too strong. To reduce:

1. Right-click node → Add Node → Parallel Layer
2. One path: original (no LUT)
3. Other path: LUT applied
4. Adjust Key Output (node opacity) to blend

Or simply:

- Apply LUT to a Layer Mixer node
- Adjust layer opacity to taste (try 30-50%)

Method 2: Manual Color Wheel Look (Most Control)

Using the color wheels to create color separation between shadows/highlights:

Classic "Orange and Teal" (Hollywood standard):

- Lift (shadows): Push toward TEAL/BLUE
- Gain (highlights): Push toward ORANGE/WARM
- Gamma (midtones): Keep neutral or very slight warm

- Amount: Very subtle - barely visible individually

Warm/Nostalgic:

- Lift: Warm (toward yellow/orange)
- Gain: Warm, slightly more than lift
- Gamma: Slight warm
- Overall saturation: Might reduce slightly

Cold/Clinical:

- Lift: Cool blue
- Gain: Cool, slightly cyan
- Gamma: Neutral
- Overall: Desaturated feeling

High Contrast Drama:

- Lift: Deep, cool blue
- Gain: Hot, slightly yellow
- Contrast: Increased
- Saturation: Often reduced in shadows, boosted in highlights

Method 3: Per-Channel Curves (Advanced)

Each RGB channel can be adjusted separately:

Blue Channel (most common for looks):

- Raise shadows = Blue shadows
- Lower highlights = Warm/yellow highlights
- This is the basis of most "cinematic" looks

Red Channel:

- Raise shadows = Magenta/warm shadows
- Lower highlights = Cyan highlights

Green Channel:

- Raise shadows = Green shadows (rarely wanted)
- Lower highlights = Magenta highlights

Classic Blue Shadows / Warm Highlights:

1. Go to Curves → dropdown → select "B" (Blue channel)
2. Add point in shadows (around 20-30%), drag UP slightly
3. Add point in highlights (around 70-80%), drag DOWN slightly
4. Result: Blue-ish shadows, warm-ish highlights

Common Look Recipes

Clean/Natural (Documentary, Corporate)

- Lift: Neutral
- Gamma: Neutral
- Gain: Neutral to very slight warm
- Saturation: Default to +5
- Intent: Just enhance reality, don't stylize

Warm/Inviting (Commercials, Rom-coms)

- Lift: Slight warm
- Gamma: Slight warm
- Gain: Warm (toward yellow-orange)
- Saturation: +5 to +10
- Intent: Everything feels sunny and welcoming

Cool/Tense (Thriller, Drama)

- Lift: Blue/teal
- Gamma: Neutral to slight cool
- Gain: Neutral to slight warm (for skin)
- Saturation: -5 to default
- Intent: Tension, unease

High Fashion/Beauty

- Lift: Neutral to slight cool
- Gamma: Bright, open
- Gain: Clean white
- Saturation: Controlled, skin neutral
- Intent: Clean, aspirational

Gritty/Desaturated (War films, Gritty drama)

- Lift: Raised (not black)
- Gamma: Flat
- Gain: Rolled off (not white)
- Saturation: -20 to -40
- Intent: Raw, documentary feel

Orange and Teal (Action, Blockbuster)

- Lift: Teal (cyan-blue)
- Gamma: Neutral
- Gain: Orange-warm
- Saturation: Boosted
- Intent: Punchy, exciting

The "Toggle Test"

This is the most important test for any creative look:

1. Toggle N9 off (Cmd/Ctrl + D)
2. Look at the image without the look
3. Toggle N9 back on
4. Ask: Does this look ADD to the story, or is it just "different"?

If the look is good:

- The image feels more emotionally appropriate with it on
- The mood matches the content
- It enhances without distracting

If the look is bad:

- The image is just "more blue" or "more orange" without purpose
- The mood doesn't match the content
- You notice the look more than the image

Applying Looks Consistently

Once you have a look you like:

Method 1: Group Clips

1. Select all clips in the scene
2. Right-click → Add Clips to Group

3. Apply look to Group Pre-Clip or Group Post-Clip grade
4. Individual clips can still have shot-specific adjustments

Method 2: Use Shared Nodes

1. In the node graph, right-click → Add Node → Shared Node
2. Apply your look to this shared node
3. Shared node syncs across all clips using it
4. Change once, changes everywhere

Method 3: PowerGrade

1. Create your look on one clip
2. Right-click thumbnail → Grab Still
3. Gallery → Right-click → Export → PowerGrade
4. Apply to other clips, adjust as needed

Common Mistakes

❌ **Adding a look before N1-N8 are done** - Style on bad foundation = garbage ❌ **Going too extreme** - If anyone says "wow, that's really teal," it's too much ❌ **Using a look that doesn't match content** - Cool look on a warm beach scene = wrong ❌ **Same look for everything** - Different scenes might need different intensities ❌ **Fighting the footage** - Work with what you shot, not against it ❌ **Forgetting about skin** - Skin should still look human under any look

Extra Nodes to Consider (N9 variants)

N9b - LUT Blend Node: To control LUT intensity properly:

```
HOW: Add Parallel structure before LUT
NODE A: Original (bypass)
NODE B: LUT applied
OUTPUT: Adjust Key Output to blend (30-50% common)
```

N9c - Film Print Emulation: For the most authentic film look:

```
TOOL: Film print emulation LUT + subtle grain
WHERE: N9 for color, N7 for grain
TIP: Real film looks are subtle - 10-30% LUT strength
```

N9d - Shot-Specific Look Adjustments: When some shots need different intensity:

```
HOW: Add node after N9 for shot-specific tweaks
USE: Day-to-night adjustment, interior vs exterior intensity
KEEP: Base look consistent, only adjust intensity
```

End State Checklist

✓ Look matches the emotional intent of the content
✓ Skin tones still look human
✓ Toggle test passes (image is better with look, not just different)
✓ Look intensity is appropriate (not obviously graded)
✓ Consistency across shots in the scene
✓ Enhances the story, doesn't distract from it

N10 — OUTPUT TRIM / LEGAL SAFE

Purpose: Ensure your final output meets technical specifications and doesn't contain any illegal values. This is your insurance policy.

```
┌──────────────────────────────────────────────────────┐
│ TOOL: Video Limiter (for broadcast) or Soft Clip     │
│                                                      │
│ VIDEO LIMITER:                                       │
│ → Effects Library → OpenFX → ResolveFX Refinement →  │
│  Video Limiter (drag onto node)                      │
│ → Set to "Soft Clip" mode, levels depend on delivery spec │
│                                                      │
│ SOFT CLIP (built-in):                                │
│ → Color page → Bottom left → Primaries Bars →        │
│  High Soft / Low Soft sliders (clip highlights/shadows) │
│                                                      │
│ CHECKING LEVELS:                                     │
│ → View your Waveform scope - nothing should exceed 100% │
│  or go below 0% (for web) or outside 16-235 (for broadcast) │
└──────────────────────────────────────────────────────┘
```

What This Node Actually Does

This node catches any values that might have crept out of legal range during your grading. It's a safety net, not a creative tool.

If your grade is good, this node does almost nothing. That's the point.

Understanding Legal Levels

Different delivery destinations have different requirements:

Web/YouTube/Vimeo (Full Range):

- Black: 0%
- White: 100%
- No illegal values exist - anything is technically "legal"
- But clipped highlights (100%) have no detail

Broadcast (Legal Range / Limited Range):

- Black: 16 (8-bit) / 64 (10-bit) = ~6%
- White: 235 (8-bit) / 940 (10-bit) = ~92%
- Values outside this range can cause problems
- Broadcast equipment may clip or behave unexpectedly

HDR (Depends on Standard):

- Different rules entirely
- Consult specific HDR specifications

Step-by-Step Workflow

Step 1: Check Your Delivery Specs

Before doing anything, know your destination:

- YouTube/Web: Full range (0-100%)
- TV broadcast: Legal range (16-235)
- Streaming (Netflix, etc.): Check their specific requirements
- Film festival: Usually full range

Step 2: Check Your Current Levels

Open your Waveform scope:

1. Look at the very top - anything hitting or exceeding 100%?
2. Look at the very bottom - anything at or below 0%?
3. Any significant amount of clipping?

Step 3: If Needed, Add Limiters

For web (minimal intervention): Usually just verify no severe clipping. If highlights are barely touching 100%, it's probably fine.

For broadcast (Video Limiter):

1. Effects Library → OpenFX → ResolveFX Refinement → Video Limiter

2. Drag onto N10 node
3. In Inspector, set:
   - Output Range: Legal (or specific values: 64-940 for 10-bit)
   - Soft Clip: ON (more natural than hard clipping)

Step 4: Final Check

After adding limiters:

1. View waveform again
2. Verify all values are now within legal range
3. Play through the timeline looking for any spikes

Soft Clip vs. Hard Clip

Hard Clip:

- Everything above the limit becomes exactly the limit value
- Abrupt, can look harsh
- Use only when specs are absolute

Soft Clip:

- Gradually compresses values approaching the limit
- More natural rolloff
- Better for maintaining perceived detail

Always prefer Soft Clip unless told otherwise.

Manual Soft Clipping (Built-in)

If you don't want to use the Video Limiter OFX:

1. Go to Primaries Bars panel
2. Find "High Soft" slider (affects highlights)
3. Pull down slightly to compress peaks
4. Find "Low Soft" slider (affects shadows)
5. Pull up slightly if needed

Common Mistakes

❌ **Relying on this node to fix bad grades** - Fix problems in earlier nodes ❌ **Using hard limits on web content** - No benefit, just clips your detail ❌ **Forgetting to check specs** - Different destinations need different limits ❌ **Not checking the final output** - Always verify after this node

Extra Nodes to Consider (N10 variants)

N10b - Color Space Conversion Out: If you need to convert to a specific delivery color space:

```
TOOL: Color Space Transform
USE: Converting from working space to delivery space
EXAMPLE: DaVinci Wide Gamut → Rec.709 for web
WHERE: Can be here or in Deliver page settings
```

N10c - Dithering (for 8-bit delivery): If delivering to 8-bit and seeing banding:

```
TOOL: Add subtle grain or noise
PURPOSE: Dithering hides banding in 8-bit outputs
AMOUNT: Very subtle - just enough to mask banding
```

End State Checklist

✓ You know your delivery specifications
✓ Waveform shows all values within legal range
✓ Soft clip is used (not hard clip) where appropriate
✓ No severe clipping of important detail
✓ Node does minimal work (grade is already good)

WORKFLOW ORDER (SHOT MATCHING) - DETAILED

Remember: Work HORIZONTALLY, not vertically. Complete each node across ALL shots before moving to the next node.

Phase 0: Setup

1. Grade your **hero shot** first (the best/most representative shot in the scene)
2. Build the complete 10-node structure on this shot
3. Save as **PowerGrade**
4. Apply the PowerGrade to all clips in the scene
5. **Grab a still** of your hero shot for reference

Phase 1: Physical Correction (N1–N4)

N2 - Primary Balance (ALL SHOTS):

- Use wipe comparison (Cmd/Ctrl + W) against your hero still
- Match exposure and white balance across all shots
- Check waveform - skin tones should land at similar heights

- This is the most important matching phase

N3 - Contrast (ALL SHOTS):

- Apply similar contrast curves across the scene
- Check that shadow and highlight roll-off matches
- Scene should feel cohesive in terms of "density"

N4 - Highlight Safety (ALL SHOTS):

- Usually similar settings across all shots
- Verify no highlight blow-out when you boost saturation later

Phase 2: Perceptual Enhancement (N5–N7)

N5 - Saturation (ALL SHOTS):

- Similar saturation levels create visual consistency
- Watch for shots that need less (already punchy) or more (flat)

N6 - Hue Sculpt (ALL SHOTS):

- Keep hue adjustments consistent within a scene
- Sky should be the same blue, grass the same green

N7 - Texture (ALL SHOTS):

- Match the "feel" across shots
- Same camera = same settings usually

Phase 3: Finishing (N8–N10)

N8 - Vignette (SHOTS THAT NEED IT):

- Not every shot needs vignette
- Apply where it helps guide the eye

N9 - Creative Look (ALL SHOTS):

- Same look applied to all = instant consistency
- Fine-tune intensity if needed per shot

N10 - Output (ALL SHOTS):

- Verify legal levels
- Final safety check

Why This Order Works

```
When you do N2 on shot 5:
- Shots 1-4 are already balanced
- You can wipe-compare against them
- Easy to match

When you do N9 on shot 5:
- Shots 1-4 already have the same look
- N9 just needs to be applied, minimal tweaking
- Everything matches automatically
```

If you finished shot 1 completely before starting shot 2, you'd be trying to match a fully-graded image to raw footage - much harder.

GOLDEN RULES

Where does this fix belong?

- Affects all shots → earlier node (N2–N4)
- Affects specific colors → N6
- Affects mood/feel → N9
- Affects one shot only → create a new node after N10

Before you "just tweak saturation":

- Stop
- Ask: What am I actually trying to fix?
- Find the right node for that job

The bypass test:

- Toggle each node off and on
- If the change is invisible, the node might be unnecessary
- If the image breaks, the node is doing too much

QUICK REFERENCE: SCOPES

```
┌──────────────────────────────────────────────────────────────┐
│ HOW TO OPEN SCOPES:          │
```

```
│ → Color page → Top right corner → Click "Scopes" button │
│ → Or: Workspace menu → Video Scopes │
│ → Or: Shortcut Cmd/Ctrl + Shift + W │
│ │
│ TO CHANGE SCOPE TYPE: Right-click on the scope → Select type │
└─────────────────────────────────────────────────────────────┘
```

Scope	Use For	Watch
Waveform (Luma)	Exposure, contrast	Clipping at 0 or 100
Waveform (RGB Parade)	Color balance, channel clipping	Matched tops/bottoms
Vectorscope	Skin tones, saturation	Skin on the skin tone line
Histogram	Overall distribution	Gaps, clipping

Skin tone line: Runs from center toward 11 o'clock on the vectorscope. All natural skin tones (any ethnicity) should fall on or very near this line.

COMMON PROBLEMS & FIXES

Problem	Likely Cause	Fix Location
Image feels flat	Insufficient contrast	N3
Skin looks orange	Over-saturated or wrong WB	N2 or N5
Sky is breaking up	Highlight saturation	N4
Shadows are noisy	Too much saturation or lift	N5 or N2
Image looks "graded"	Over-styled	N9 (reduce)
Colors are unnatural	Hue shifts too aggressive	N6
Harsh/digital texture	Needs midtone detail	N7
Shot doesn't match	Primary balance off	N2
Halos around objects	Over-sharpened	N7 (reduce Mid Detail)
Banding in gradients	Too much contrast or 8-bit	N3 (reduce) or add grain in N7

POWERGRADE TEMPLATE SETUP

Getting to the Color page:

- Press Shift + 6, or click "Color" in the bottom toolbar

To save this workflow as a template:

1. Build the complete 10-node structure
2. Label each node clearly (N1, N2, etc. or full names)
3. Right-click the thumbnail → Grab Still
4. In the Gallery, right-click → Export → PowerGrade

To apply:

1. Right-click in Gallery → Import → PowerGrade
2. Right-click the grade → Apply Grade
3. Adjust values per shot (structure stays)

PODCAST-SPECIFIC WORKFLOW

For talking-head podcast content, you'll likely need additional techniques beyond the standard 10-node workflow. This section covers three common podcast needs:

1. Separating subject from background (brighten face, darken background)
2. Cosmetic enhancement (reducing wrinkles, under-eye circles)
3. Grading full source clips before cutting (so tracking works across the entire recording)

PODCAST NODE STRUCTURE

For podcast work, insert these additional nodes after your standard workflow:

```
N1-N10: Standard workflow (as documented above)
N11: Subject/Background Separation (Magic Mask)
N12: Face Enhancement (Face Refinement or Beauty)
N13: Under-Eye Fix (if needed)
N14: Eye Enhancement (if needed)
```

P1 — SEPARATING SUBJECT FROM BACKGROUND

Purpose: Brighten the face/subject while darkening or desaturating the background to make the speaker pop.

```
┌──────────────────────────────────────────────────────────────┐
│ TOOL: Magic Mask (AI-powered subject isolation)              │
│ LOCATION: Color page → Bottom center panel → Magic Mask icon │
│          (between Blur and Tracker icons)                    │
│                                                              │
│ ALTERNATIVE: Power Window with tracking                      │
│ LOCATION: Color page → Window tab → Circle/Polygon           │
└──────────────────────────────────────────────────────────────┘
```

Method 1: Magic Mask (Easiest, Studio Version Only)

Magic Mask uses AI to automatically detect and track people in your footage.

Step-by-Step:

1. **Create a new serial node** after your main grade (Alt/Option + S)
2. **Click Magic Mask icon** in the center toolbar
3. **Select "Person"** tab (or "Features" if you only want the face)
4. **Click "Add" tool** (plus icon)
5. **Draw a stroke** through your subject's torso/head - just a quick line, not an outline
6. The AI will automatically detect and mask the person (shown in red overlay)
7. **Toggle Mask Overlay** (icon looks like a square) to see what's selected
8. **Track the mask:** Click the forward/backward tracking buttons to track through the clip
9. Now any adjustments on this node only affect the person

To affect BACKGROUND instead:

- Click **"Invert"** button in the Magic Mask panel
- Now adjustments affect everything EXCEPT the person

Typical Background Adjustments (on inverted mask):

- Lower exposure (Offset down -0.15 to -0.30)
- Reduce saturation slightly
- Optional: Add slight blur for depth-of-field effect

Typical Subject Adjustments (on non-inverted mask):

- Slight lift to exposure if needed
- Gentle contrast boost

- Keep saturation natural

Method 2: Power Window with Face Tracking (Works in Free Version)

If you don't have Studio, or Magic Mask isn't working well:

1. **Create a new serial node**
2. **Go to Window tab** (bottom center panel)
3. **Select Circle or Polygon** shape
4. **Draw around your subject's face/upper body**
5. **Add heavy softness** (Soft 1-4 sliders, crank them up)
6. **Go to Tracker tab**
7. **Select "Window"** tracking type
8. **Click Track Forward/Backward** to track the window
9. Apply adjustments to the subject

To affect background: Click "Invert" in the Window panel.

Podcast-Specific Tips

For stationary talking heads:

- Tracking is usually simple since the subject doesn't move much
- You can often get away with a static oval window with heavy feathering
- Check beginning, middle, and end of clip for drift

For multiple people in frame:

- Use Magic Mask → Features → Face to isolate just faces
- Or create separate nodes with separate masks for each person

Performance note: Magic Mask can be slow on long clips. Consider:

- Rendering cache (right-click clip → Render Cache → User)
- Or work with proxies during editing

Reusing Magic Mask Tracking for Multiple Nodes (NO RE-TRACKING!)

If you've already tracked a Magic Mask and want separate nodes for subject AND background without running the track again:

```
┌──────────────────────────────────────────────────────────┐
│ THE 10-SECOND FIX: Route the Alpha (mask) to a new node  │
│                                                          │
│ Your Magic Mask node has a Blue Square output (Alpha Output) │
│ You can connect this to another node's Blue Triangle input │
│ (Alpha Input) to share the mask data without re-tracking │
└──────────────────────────────────────────────────────────┘
```

Step-by-Step:

1. **Create a new Serial Node:** Press Alt/Option + S to create a new, empty node next to your Magic Mask node. This will be your "Background" node.
2. **Connect the Alpha (the blue line):**
   - Look at your Magic Mask node - it has a **Blue Square** on the right side (Alpha Output)
   - Look at your new node - it has a **Blue Triangle** on the left side (Alpha Input)
   - **Drag a line** from the Blue Square to the Blue Triangle
   - Visual check: You should now have a **dotted blue line** connecting the two nodes
3. **Invert the new node:**
   - Select the new node (the Background one)
   - Go to the **Key** palette (icon looks like a key, usually 5th icon in the center tool strip)
   - Find the **Key Output** section
   - Click the **Invert** icon (looks like a white circle inside a black square)

Result:

```
[Magic Mask Node] ════════════════════════════════▶ Affects PERSON only
       │
       │ (blue dotted line - alpha connection)
       ▼
[New Node with Inverted Key] ═══════════════════════▶ Affects BACKGROUND only
```

Both nodes use the SAME tracking data. Zero additional render time.

Pro Tip - Protect Your Work: Since long Magic Mask tracks can take an hour+, protect yourself:

- **Right-click the clip** in the timeline (Edit page)
- Select **Render in Place** (choose a high-quality codec like DNxHR or ProRes)
- This "bakes" the grade so you never lose the Magic Mask cache if Resolve crashes or you accidentally click the wrong button

---

P2 — COSMETIC ENHANCEMENT (FACE REFINEMENT)

Purpose: Subtly improve skin appearance - reduce wrinkles, smooth skin, brighten eyes - without looking "filtered."

```
┌──────────────────────────────────────────────────────────────┐
│ TOOL: Face Refinement (OpenFX plugin - Studio only)          │
│ LOCATION: Effects Library → OpenFX → ResolveFX Refine →      │
│         Face Refinement (drag onto a node)                   │
│                                                              │
│ ALTERNATIVE: Beauty / Midtone Detail + Power Windows         │
│ LOCATION: Primaries Bars → Mid/Detail slider (negative values) │
│         Combined with Window to isolate face                 │
└──────────────────────────────────────────────────────────────┘
```

Method 1: Face Refinement Plugin (Recommended for Podcasts)

Face Refinement automatically tracks faces and gives you per-feature controls.

Step-by-Step:

1. **Create a new serial node** after subject separation (if used)
2. **Open Effects Library** (top left, or Workspace → Effects Library)
3. **Search "Face Refinement"** or navigate: OpenFX → ResolveFX Refine → Face Refinement
4. **Drag onto your node**
5. **In the Inspector (top right), click "Analyze"** - wait for it to track the face
6. You should see tracking markers appear on the face in the viewer
7. If not, enable **OpenFX Overlay** in the viewer dropdown

Recommended Settings for Podcasts:

Section	Setting	Typical Value	Notes
Texture	Mode	Beauty Automatic	Start here, switch to "Smoothing" for more control
Texture	Amount	0.3 - 0.6	Higher = smoother. Around 0.5-0.6 is good medium. Don't exceed 0.7 or it looks plastic
Texture	Scale	0.3 - 0.5	Controls how much of the face area is affected. Work both sliders together
Eye Retouching	Sharpen	0.1 - 0.2	Very subtle, keeps eyes crisp
Eye Retouching	Brighten	0.1 - 0.15	Subtle lift to eye area
Color Grading	Contrast	0.05 - 0.1	Slight face contrast boost
Forehead	Smooth	0.1 - 0.3	Reduces forehead lines
Cheek	Smooth	0.1 - 0.2	Reduces cheek lines

Quick Starting Points by Desired Effect:

Look	Amount	Scale	Notes
Subtle/natural	0.25 - 0.35	0.3 - 0.4	Barely noticeable, professional
Medium (most podcasts)	0.4 - 0.55	0.4 - 0.5	Good balance
Heavy (beauty/makeup content)	0.55 - 0.7	0.5 - 0.6	More obvious smoothing

How to Dial It In:

1. Start with Amount at **0.4** and Scale at **0.4**
2. Push Amount up until it looks too smooth/plastic
3. **Back off 25%** from wherever it started looking fake
4. Adjust Scale to cover more or less of the face area
5. Toggle bypass (Cmd/Ctrl + D) to compare before/after

Critical: Less is more. If someone can tell you've applied face refinement, you've gone too far. The goal is "they look good on camera," not "they've been retouched."

Method 2: Midtone Detail + Window (Works in Free Version)

If you don't have Face Refinement:

1. **Create a new serial node**
2. **Add a Power Window** (oval) around the face
3. **Track it** using the Tracker panel
4. **Add heavy softness** to the window edges
5. **Go to Primaries Bars**
6. **Set Mid/Detail to -15 to -25** (negative = softer)

This softens skin texture without a dedicated face tool. It's less precise but works.

What Each Face Refinement Section Does

Texture tab:

- Smoothes skin while (theoretically) preserving detail
- "Beauty Automatic" is the easiest mode
- "Smoothing" mode gives you more control with Detail/Smoothing sliders

Eye Retouching tab:

- Sharpen: Increases perceived detail in eyes
- Brighten: Lifts the eye area slightly
- Important: Keep eyes sharp while softening skin

Lip Retouching tab:

- Can add color/saturation to lips
- Usually leave alone for podcast/natural look

Forehead/Cheek/Chin tabs:

- Per-region smoothing control
- Useful if forehead has lines but cheeks are fine

Blush tab:

- Adds artificial blush color to cheeks
- Usually not needed for podcasts (too "made up" looking)

Color Grading tab:

- Affects just the face

- Contrast, Tint, Saturation just for the isolated face

P3 — FIXING UNDER-EYE CIRCLES (PANDA EYES)

Purpose: Reduce dark circles under eyes without looking unnatural.

```
┌──────────────────────────────────────────────────────────────┐
│ TOOL: Custom Power Window + Contrast/Exposure adjustment      │
│ LOCATION: Color page → Window tab → Pen tool (custom shape)   │
│          or Circle tool positioned under each eye             │
│                                                               │
│ CONCEPT: Emulate fill light by reducing contrast in dark areas │
└──────────────────────────────────────────────────────────────┘
```

The Technique (Manual but Effective)

Under-eye circles are essentially shadows. The fix is to emulate what a fill light would do - lift the shadows without affecting highlights.

Step-by-Step:

1. **Create a new serial node** after Face Refinement
2. **Go to Window tab**
3. **Select the Pen tool** (for kidney-bean shapes) or use two small **ovals**
4. **Draw a shape under one eye** - kidney-bean or crescent shape following the dark area
5. **Add heavy softness** (Soft sliders to 70-100%) - NO visible edges
6. **Track the window** (Tracker tab → Track Forward/Backward)
7. **Repeat for the other eye** (or use one larger shape covering both)

Now make the adjustment:

8. **Reduce Contrast** (pull down to -10 to -25) - This lifts shadows while pulling down highlights, reducing the darkness
9. **Slight Gamma lift** if needed (push up +0.05 to +0.10) - Don't overdo it
10. **Check color:** Reduced contrast can make the area look gray/desaturated
11. **Add slight warmth or saturation** to match surrounding skin if needed

Why Reduce Contrast Instead of Just Lifting?

- Lifting exposure alone brightens everything including the highlights, looking unnatural
- Reducing contrast compresses the tonal range, bringing shadows up AND highlights down
- This emulates what fill light actually does - it doesn't brighten, it reduces shadow depth

Common Mistakes

❌ **Hard-edged windows** - Creates obvious patches under eyes ❌ **Lifting too much** - Under-eye becomes brighter than surrounding skin ❌ **Forgetting to color match** - Fixed area looks gray compared to skin ❌ **Over-smoothing with blur** - Looks like concealer applied by a child

Alternative: Use Face Refinement's Eye Section

Face Refinement has an "Eye Retouching → Brighten" parameter that can help with minor dark circles, but it's less targeted than a custom window.

P4 — GRADING FULL SOURCE CLIPS (BEFORE CUTTING)

Purpose: When you have a pre-cut podcast edit, and you need face tracking or grades to apply to the FULL recording (not just the cut portions).

```
┌─────────────────────────────────────────────────────────┐
│ THE PROBLEM:                                            │
│ You've edited a 1-hour podcast into 20 cuts from the same clip. │
│ Face tracking only tracks within each cut segment.      │
│ If you track on the cut timeline, each segment tracks   │
│ independently and may not match.                        │
│                                                         │
│ THE SOLUTION:                                           │
│ Grade the FULL source clip first, THEN edit (or re-conform). │
└─────────────────────────────────────────────────────────┘
```

Method 1: Grade First, Edit Second (Best Workflow)

If you're starting a new podcast project:

1. **Import your full recording** into Media Pool
2. **Create a "grading timeline"** with the full, uncut clip
3. **Do all your grading, face tracking, face refinement** on this full clip
4. The tracking runs across the entire recording seamlessly
5. **Grab a still** of your grade and save as PowerGrade
6. **Now create your edited timeline** and do your cuts
7. The grade from the source clip carries over to all cuts

Why this works: When you grade a source clip, the grade is attached to the clip itself. All instances of that clip in any timeline inherit the grade.

Method 2: Remote Grades (For Already-Edited Timelines)

If you've already edited and need to grade:

1. **In Color page,** right-click on your clip
2. **Enable "Use Remote Grades"** (or find it in the Color menu)
3. Grade one instance of the clip
4. **All other instances of the same source clip** automatically get the same grade

Limitation: Remote grades share the SAME grade. If you need different grades for different sections, this won't work.

Method 3: Create a Full-Clip Timeline for Tracking

If you need face tracking across the full source:

1. **Create a NEW timeline** (File → New Timeline)
2. **Drag the FULL source clip** (uncut) into this timeline
3. **Apply your grade with face tracking/refinement** to this full clip
4. Track the entire duration - Magic Mask or Face Refinement will track continuously
5. **Grab a still** of the grade
6. **Go back to your edited timeline**
7. **Apply the grade** from the still to your cut clips

The tracking data is baked into the grade from the full clip, so when you apply it to cuts, the tracking should align (since they're from the same source at the same timecodes).

Method 4: Group Pre-Clip for Consistent Base Grade

If your podcast uses the same source clip cut multiple times:

1. **In Color page, select all clips** from that source (use thumbnail view)
2. **Right-click → Add to New Group**
3. **Change Node Editor mode** from "Clip" to "Group Pre-Clip"
4. **Apply your base grade here** (no tracking - just color/exposure)
5. This grade affects ALL clips in the group before individual clip grades

For tracking: You'll still need to track each segment individually, but at least the base grade is unified.

The Harsh Reality of Podcast Tracking

Truth: If you've already cut your timeline into many small pieces, and you need face tracking that spans the cuts, there's no perfect solution in Resolve.

Best practices:

- Grade and track the FULL source BEFORE editing whenever possible
- For already-cut timelines, you may need to track each segment individually
- Use Groups for consistent base grades
- Accept that tracking may need touchup at cut points

PODCAST QUICK REFERENCE

Task	Tool	Location
Isolate person from background	Magic Mask	Color page → Magic Mask icon
Darken background only	Magic Mask (inverted)	Magic Mask → Invert button
Smooth skin	Face Refinement or Mid/Detail	Effects → Face Refinement, or Primaries → Mid/Detail
Brighten eyes	Face Refinement	Face Refinement → Eye Retouching → Brighten
Sharpen eyes	Face Refinement	Face Refinement → Eye Retouching → Sharpen
Fix under-eye circles	Power Window + Contrast	Window → Pen tool, then reduce Contrast
Grade full source before cuts	Create grading timeline	New Timeline with full source clip
Apply same grade to all cuts	Remote Grades or Groups	Right-click → Use Remote Grades, or Group Pre-Clip

---

PODCAST NODE TREE EXAMPLE

```
N1:  Input Transform (if shooting LOG)
N2:  Primary Balance (exposure, white balance)
N3:  Contrast Shaping
N4:  Highlight Safety
N5:  Saturation
N6:  Hue Sculpt
N7:  Texture (midtone detail)
N8:  (skip for podcasts - vignette not usually needed)
N9:  Creative Look (optional)
N10: Output Safety
────────────────────────────────────────
N11: Background Darken (Magic Mask inverted, reduce exposure)
N12: Subject Enhance (Magic Mask normal, slight lift/contrast)
```

```
N13: Face Refinement (smooth skin, brighten eyes)
N14: Under-Eye Fix (Power Window, reduce contrast)
```

Order matters: Do cosmetic work AFTER color grading, so your face refinement is working on correctly balanced footage.

RENDERING, CACHING, AND PERFORMANCE

This section explains the different ways to pre-render footage in Resolve, when to use each, and how to protect expensive work like Magic Mask tracking.

UNDERSTANDING THE THREE OPTIONS

```
┌──────────────────────────────────────────────────────┐
│ RENDER CACHE vs RENDER IN PLACE vs FINAL RENDER       │
│                                                       │
│ These are THREE DIFFERENT THINGS with different purposes │
└──────────────────────────────────────────────────────┘
```

Option	What It Does	Files Created	Portability	Use Case
Render Cache	Pre-renders for smooth playback	.dvcc files in cache folder	LOCAL ONLY - can't share	Day-to-day editing
Render in Place	Bakes effects into new media file	Actual video files (ProRes/DNxHR)	PORTABLE - can share	Protecting work, collaboration
Final Render	Export from Deliver page	Final deliverable	PORTABLE	Client delivery

---

RENDER CACHE (For Smooth Playback)

What it is: Resolve pre-renders clips in the background so playback is smooth.

Location: Playback → Render Cache → Smart or User

```
┌─────────────────────────────────────────────────────────────┐
│ SMART MODE: Resolve automatically decides what to cache     │
│ - RED line = needs caching                                  │
│ - BLUE line = cached, will play smoothly                    │
│                                                             │
│ USER MODE: You manually choose what to cache                │
│ - Right-click clip → Render Cache Color Output → On         │
└─────────────────────────────────────────────────────────────┘
```

Key Points:

- Cache files are .dvcc format - ONLY usable in Resolve, ONLY on your machine
- If you clear cache or move project, you lose the cache
- Cache is NOT used in final export by default
- Good for: day-to-day smooth editing
- Bad for: protecting expensive work (like hour-long Magic Mask tracks)

Cache Format Setting: Project Settings → Master Settings → Optimized Media and Render Cache → Render Cache Format

Format	When to Use
DNxHR SQ	General work, good balance of quality/size
DNxHR HQX (10-bit)	Better quality, larger files
ProRes 422 HQ	Mac users, good quality
ProRes 4444 / DNxHR 444	If you need to preserve alpha channels

WARNING: Render Cache can be FRAGILE. Magic Mask data is stored in cache. If cache is cleared, corrupted, or project is moved, Magic Mask may need to re-track. This is why Render in Place exists.

RENDER IN PLACE (For Protecting Work)

What it is: Creates actual video files with effects "baked in" that replace clips on timeline.

Location: Edit page → Select clip(s) → Right-click → Render in Place

```
┌─────────────────────────────────────────────────────────┐
│ RENDER IN PLACE creates REAL VIDEO FILES you can:        │
│ - Move to another computer                               │
│ - Share with collaborators                               │
│ - Keep forever as backup                                 │
│                                                          │
│ REVERSIBLE: Right-click → Decompose to Original          │
│ (restores original clip with all effects still there)    │
└─────────────────────────────────────────────────────────┘
```

Render in Place Options:

Option	Recommendation
Include Video Effects	✅ Yes - bakes Edit page effects
Include Fusion Composition	✅ Yes - bakes Fusion work
Include Color Grading Effects	⚠️ CAREFUL - see below
Add Handles	✅ Yes, 10 frames - gives flexibility
Use Source Resolution	✅ Yes - preserves quality

The "Include Color Grading" Decision:

- ✅ **CHECK IT** if you want to permanently bake your grade (no more changes)
- ❌ **UNCHECK IT** if you want to grade ON TOP of the rendered file (keep flexibility)

For podcast Magic Mask work: Usually UNCHECK color grading, so you can still adjust colors after baking the mask.

Codec Choice for Render in Place:

Platform	Standard Work	With Alpha Channel
Mac	ProRes 422 HQ	ProRes 4444
Windows/Linux	DNxHR HQX (10-bit)	DNxHR 444 (12-bit)

THE MAGIC MASK PROTECTION PROBLEM

The situation: You spent 1 hour tracking a Magic Mask. That tracking data lives in the cache. If anything goes wrong with the cache, you lose an hour of work.

Face Refinement vs Magic Mask - Tracking Persistence

Tool	Where Tracking is Stored	Survives Project Close/Reopen?	Survives Cache Clear?
Face Refinement	Project database	✅ YES	✅ YES
Magic Mask	Cache folder	✅ YES (if cache intact)	❌ NO - must re-track

Face Refinement: When you click "Analyze", the tracking data is stored in the project database. Close the project, reopen it, the tracking is still there. You only need to re-analyze if you change the clip itself.

Magic Mask: The tracking is stored in the cache folder (those .dvcc files). If you:

- Clear render cache
- Move project to another computer
- Cache gets corrupted
- Cache drive fails

...you lose the Magic Mask tracking and must re-track from scratch.

Conclusion: Face Refinement is relatively safe. Magic Mask needs protection via Render in Place or External Matte export.

PROTECTING MAGIC MASK: SIMPLE METHOD (Render in Place)

Use this if: You just want to bake everything and move on. You won't need to independently adjust person/background later.

Your Current Node Setup

```
[Source] → [Node 1: Magic Mask - PERSON adjustments]
                  ↓ (blue alpha line)
           [Node 2: BACKGROUND adjustments - inverted key]
```

Step-by-Step Instructions

Step 1: Go to Edit Page

- Press **Shift + 4** or click **Edit** at bottom of screen

Step 2: Select your clip

- Click on the clip that has Magic Mask applied
- Should be your FULL SOURCE clip (not cuts)

Step 3: Right-click → Render in Place

Step 4: Configure settings

```
┌──────────────────────────────────────────────────────────────┐
│ Format: QuickTime                        │
│ Codec: Apple ProRes 422                  │
│                                          │
│ ☑ Include Video Effects       ← CHECK              │
│ ☑ Include Fusion Composition    ← CHECK              │
│ ☑ Include Color Grading Effects  ← CHECK (bakes mask + grade) │
│                                          │
│ ☑ Add Handles: 0 frames (or 10 for safety)           │
│ ☑ Use Source Resolution                  │
└──────────────────────────────────────────────────────────────┘
```

Step 5: Click Render

- Choose a folder to save the file
- Name it something like `Source_Graded_Baked.mov`

Step 6: Wait for completion

- Progress bar should move steadily
- If it's slow/stuck, see Troubleshooting section below
- When done, timeline clip is replaced with rendered file

Step 7: Done!

- Your timeline now uses the pre-baked file
- Magic Mask never needs to run again
- You can clear cache, move project - the grade is permanent
- Your CUT timeline (if separate) will also use this baked file automatically

If You Need to Change Something Later

1. **Right-click the clip → Decompose to Original**
2. This restores original clip with all nodes intact
3. Make your changes
4. Render in Place again

PROTECTING MAGIC MASK: FLEXIBLE METHOD (External Matte)

Use this if: You want to independently adjust person and background brightness/color FOREVER without re-tracking.

Overview

You're going to:

1. Export ONLY the mask as a black & white video
2. Delete the Magic Mask node
3. Use the B&W video as a "stencil" for your adjustments

Step 1: Make the Mask Visible as Black & White

Your Magic Mask creates an invisible "alpha channel". You need to convert it to visible B&W.

1a. Add a new Serial Node after Magic Mask

- Select your Magic Mask node
- Press **Alt + S** (Windows) or **Option + S** (Mac)
- New empty node appears

1b. Connect the Alpha to RGB

```
Your Magic Mask node has these outputs:

┌─────────────────┐
│ Magic Mask  │──▶ GREEN (RGB image)
│   Node    │──▶ BLUE SQUARE (alpha/mask) ← THIS ONE
└─────────────────┘


Your new node has these inputs:

    ┌─────────────────┐
GREEN ──▶│  New Node  │
BLUE ──▶│       │
    └─────────────────┘
```

- **Click the BLUE SQUARE** on Magic Mask node
- **Drag to the GREEN TRIANGLE** on the new node
- Release

1c. Check the viewer

- You should now see BLACK & WHITE image
- WHITE = person (where mask is)

- BLACK = background

If you don't see B&W, try:

- On the new node, go to **Key tab**
- Set **Key Output Gain** to 1.0

Step 2: Disable All Color Grading

- Click the **number** on each grading node to bypass (node turns gray)
- Only Magic Mask + B&W node should be active
- Viewer should show ONLY the black & white mask

Step 3: Render in Place (Matte Only)

1. Go to **Edit page** (Shift + 4)
2. Select the clip
3. **Right-click → Render in Place**
4. Settings:

```
Format: QuickTime
Codec: ProRes 422 (or H.264 - matte doesn't need quality)

☑ Include Color Grading Effects ← YES
```

5. Click **Render**
6. Save as `Matte_Person.mov`
7. Wait for completion

Step 4: Restore Your Original Clip

1. **Right-click the clip → Decompose to Original**
2. Your original clip returns with all nodes
3. **Re-enable your grading nodes** (click numbers to un-bypass)
4. **DELETE the B&W matte node** you created (don't need it anymore)
5. **DELETE the Magic Mask node** (the matte file replaces it)

Step 5: Link the Matte to Your Source Clip

1. In **Media Pool**, find your original source clip
2. **Right-click → Clip Attributes**
3. Go to **Mattes** tab

4. Click **+ (Add)** or **Add Matte**
5. Browse to your `Matte_Person.mov` file
6. Click **OK**

Step 6: Use the Matte in Your Nodes

For PERSON adjustments:

1. Create a node (or use existing)
2. Go to **Key tab** (bottom center panel)
3. In the **Qualifier/Matte** section, click the dropdown
4. Select **Matte 1** (your linked matte)
5. Adjustments on this node now ONLY affect the person (white areas)

For BACKGROUND adjustments:

1. Create another node
2. Go to **Key tab**
3. Select **Matte 1**
4. Click **Invert** button
5. Adjustments now ONLY affect background (black areas)

Step 7: Done!

Your new node structure:

```
[Source] → [Person Node + External Matte] → [Background Node + Inverted Matte] → [Output]
```

- No Magic Mask node anywhere
- Matte is a permanent video file
- Full flexibility to adjust person/background independently
- Works in cut timeline too (timecode aligns automatically)

WHICH METHOD SHOULD YOU USE?

Situation	Use This Method
Just want it done, don't care about flexibility	**Render in Place (Simple)**
Will need to tweak person/background later	**External Matte (Flexible)**

Situation	Use This Method
Deadline tomorrow	**Render in Place (Simple)**
Client might request changes	**External Matte (Flexible)**

My recommendation for most podcast work: Just use Render in Place (Simple). If client wants changes, Decompose to Original, adjust, re-render. It's fast enough.

TROUBLESHOOTING: RENDER STUCK OR SLOW

QUICK FIX: RENDER IN PLACE (Simplest Method)

When to use: You're stuck, render won't complete, you just want this done.

What it does: Bakes your Magic Mask + grade into a permanent video file. Fast, simple, works.

Downside: Grade is baked. You can't separately adjust person vs background anymore (unless you Decompose to Original).

Step-by-Step Instructions

```
STEP 1: Cancel any stuck render
    - Click "Stop" on the dialog

STEP 2: Go to Edit Page
    - Press Shift + 4 (or click "Edit" at bottom)

STEP 3: Make sure you're on the FULL SOURCE timeline
    - The one with the complete uncut clip where you tracked the Magic Mask
    - NOT the cut timeline with 50 edits

STEP 4: Select your clip
    - Click on it so it's highlighted (orange border)

STEP 5: Right-click the clip → "Render in Place"
    - A dialog box appears

STEP 6: Set these EXACT settings:

    ┌──────────────────────────────────────────┐
    │ Format: QuickTime                  │
    │ Codec:  Apple ProRes 422           │
    │                                    │
    │ ☑ Include Video Effects    ← CHECK │
```

```
   │ ☑ Include Fusion Composition  ← CHECK     │
   │ ☑ Include Color Grading Effects ← CHECK   │
   │                                 │
   │ Add Handles: 0 frames                   │
   │ ☑ Use Source Resolution      ← CHECK     │
   └─────────────────────────────────────────┘

STEP 7: Click "Render"
   - Choose a folder (your project folder is fine)
   - Name it: Podcast_Graded_Master.mov

STEP 8: Wait for render
   - This should show actual progress
   - MUCH faster than Deliver page (uses cached tracking)

STEP 9: Done!
   - Resolve replaces your timeline clip with the rendered file
   - Magic Mask is now baked into the video
   - No more tracking computation ever
```

After Render in Place

- Your timeline now uses the rendered file
- If your CUT timeline references the same source, it automatically uses the rendered version
- Playback is instant (no Magic Mask computation)
- Final export from Deliver page will be fast

If You Need to Change Something Later

1. Select the rendered clip in timeline
2. Right-click → **"Decompose to Original"**
3. This restores the original clip with all your nodes still intact
4. Make changes
5. Render in Place again

Option 1: Render in Place (Simplest)

Just bake everything:

1. Go to Edit page
2. Select your clip with Magic Mask
3. Right-click → Render in Place

4. Format: QuickTime / DNxHR HQX
5. ✅ Include Color Grading Effects (bakes the mask effect)
6. Render

Result: Your graded clip with subject/background separation is now a permanent video file.

Downside: Grade is baked. Can't change subject brightness vs background anymore.

Option 2: Render the Matte Only (Most Flexible - Professional Workflow)

Export ONLY the mask as a black & white video, then use it as an External Matte. This is the professional approach because it permanently preserves your tracking work.

Why this is better than copy-pasting grades:

- Copy-pasted Magic Mask nodes still need cache or will re-track
- External Matte is a permanent video file - no computation ever again
- Survives cache clears, project moves, computer changes
- Final renders are instant (no AI mask computation)

Step 1: Create a B&W Matte Output Node

On your FULL SOURCE timeline (where you tracked the Magic Mask):

```
Current structure:
[Source] → [Magic Mask Node] → [Person Grade] → [Output]
              ↓ (alpha)
        [Background Node]

What you're adding:
[Source] → [Magic Mask Node] → [NEW: Matte Viewer Node] → [Output]
```

How to route the alpha (mask) to visible RGB:

1. Add a new Serial Node after your Magic Mask node (Alt/Option + S)
2. Look at your Magic Mask node - it has a **Blue Square** (alpha output) on the right
3. Look at the new node - it has a **Green Triangle** (RGB input) on the left
4. **Drag a line** from the Blue Square to the Green Triangle
5. The viewer now shows the mask as visible black & white:
   - WHITE = person (where mask is)
   - BLACK = background (where mask isn't)

Alternative method (if direct routing is tricky):

1. On the new node, go to **Key tab** (bottom center)

2. In **Key Output** section, set **Key Output Gain** to 1.0
3. The mask becomes visible in the viewer

Step 2: Export ONLY the Matte

1. **DISABLE all color grading nodes -** click the number on each node to bypass (node turns gray)
2. Only the Magic Mask + your B&W matte node should be active
3. Go to **Deliver** page
4. Settings:
   - Format: QuickTime
   - Codec: ProRes 422 (or H.264 - matte doesn't need high quality, it's just B&W shapes)
   - Filename: `Matte_Master.mov`
5. Render

Result: You now have a video file showing a white silhouette on black background, perfectly tracking your subject for the entire source clip duration.

Step 3: Link the Matte to Your Source Clip

In your **CUT timeline** (the edited one with 50 cuts):

1. **Import** `Matte_Master.mov` into your Media Pool
2. In Media Pool, find your **original source clip** (the one all your cuts come from)
3. **Right-click the source clip** → **Clip Attributes** → **Mattes** tab
4. Or: Right-click source clip → **Add Matte From** → Select `Matte_Master.mov`

This tells Resolve: "This matte file belongs to this source clip"

Step 4: Use the External Matte in Color Nodes

Now on any clip in your cut timeline:

1. Go to **Color page**
2. Create a node for person adjustments
3. Go to **Key tab** (bottom center panel)
4. Click the **Matte** dropdown → Select your linked matte
5. Adjustments on this node now only affect the WHITE areas (person)

For background:

1. Create another node
2. Add the same External Matte
3. Click **Invert** in the Key tab

4. Adjustments now only affect the BLACK areas (background)

```
New Node Structure:
[Source] → [Balance] → [Person Node + Matte] → [Background Node + Inverted Matte] → [Look] → [Output]
```

How Timecode Alignment Works:

Your cuts reference specific timecodes from the source. When you link the matte:

- Cut shows frame 00:01:30:00 of source
- Resolve automatically uses frame 00:01:30:00 of matte
- Perfect alignment, no manual syncing needed

The Magic Mask never runs again. The matte file does all the work permanently.

Option 3: Render with Alpha (For Compositing)

Export the person with transparency, layer over original for background control.

1. **On your Magic Mask timeline:**
   - Right-click in node area → Add Alpha Output
   - Connect your Magic Mask node's blue square to the blue Alpha Output dot
2. **Render in Place:**
   - Format: QuickTime
   - Codec: **ProRes 4444** or **DNxHR 444** (MUST support alpha)
   - ✅ Check "Export Alpha"
3. **In your cut timeline:**
   - Put rendered clip (person with transparency) on Video Track 2
   - Put original footage on Video Track 1 (background)
   - Grade each track independently

Downside: Huge file sizes (4444 codecs are large).

WHICH TIMELINE TO RENDER?

Scenario: You have a 1-hour podcast. You've cut it into 50 clips on your edit timeline. You tracked Magic Mask on a separate "full source" timeline.

Answer: Render the FULL SOURCE timeline, not the cut one.

Why:

1. **Continuous tracking:** The Magic Mask tracked the whole hour continuously. If you render 50 separate cuts, each would need its own tracking initialization.
2. **Simpler files:** One big "Master_Graded.mov" is cleaner than 50 separate files.
3. **Cut alignment:** The rendered full clip will have the same timecode as your source, so your cuts still reference the correct frames.

Workflow:

```
1. Full Source Timeline:
  - Apply Magic Mask
  - Track (1 hour...)
  - Apply base grade
  - Render in Place (or render matte)

2. Cut Timeline:
  - Your cuts now reference the rendered file (or use the rendered matte)
  - Everything plays smoothly
  - Magic Mask never needs to run again
```

THE "BAKING TRAP" WARNING

If you Render in Place with ALL effects including color grading:

- Person brightness and background brightness become ONE video layer
- You can no longer adjust them independently
- The flexibility is gone

If you want to keep flexibility:

- Render ONLY the matte (Option 2 above)
- Or render with alpha (Option 3 above)
- Or render in place WITHOUT color grading, then grade the rendered file

---

CODEC SPEED COMPARISON

Why is H.264 so slow to render?

H.264/H.265 are **delivery codecs** - highly compressed, small files, slow to encode.

Codec	Encode Speed	File Size	Use Case
H.264	🐌 SLOW	Small	Final delivery only
H.265/HEVC	🐌 VERY SLOW	Smaller	Final delivery only
ProRes 422 HQ	🚀 FAST	Large	Intermediate/editing
DNxHR HQX	🚀 FAST	Large	Intermediate/editing
ProRes 4444	🚀 Fast	Very Large	When you need alpha

Rule: Use ProRes/DNxHR for anything intermediate. Only use H.264/H.265 for final client delivery.

RENDER SETTINGS QUICK REFERENCE

For Render Cache (Project Settings):

```
Format: DNxHR SQ (or ProRes 422 on Mac)
Location: Fast SSD, NOT your system drive
```

For Render in Place (to protect Magic Mask):

```
Format: QuickTime
Codec: DNxHR HQX (or ProRes 422 HQ on Mac)
Include Video Effects: ✅
Include Fusion: ✅
Include Color Grading: ❌ (unless you want to bake it)
Add Handles: 10 frames
Use Source Resolution: ✅
```

For Matte Export:

```
Format: QuickTime or MP4
Codec: DNxHR SQ or even H.264 (matte doesn't need quality)
Resolution: Can be lower than source if needed
```

For Final Delivery:

```
Format: MP4
Codec: H.264 or H.265
Quality: Restrict to 20-35 Mbps for web, higher for broadcast
```

CLOUD GPU RENDERING?

Is it worth it? For one-off projects, probably not worth the setup time.

When it makes sense:

- Regular high-volume work
- Extremely long renders (feature films)
- Your local hardware is severely underpowered

Simpler alternatives:

1. **Render overnight** - start render before bed
2. **Use proxy workflow** - edit with proxies, render full res only at end
3. **Render intermediate ProRes** - fast to create, then transcode to H.264 separately
4. **Timeline Proxy Mode** - Playback → Timeline Proxy Mode → Half/Quarter resolution for editing

---

RENDER TROUBLESHOOTING

PROBLEM: "AI Magic Mask - Tracking..." Dialog Stuck (No Progress)

Symptom: The dialog says "Tracking..." but the progress bar hasn't moved for 10+ minutes.

What happened: Resolve lost the cached tracking data and is trying to re-track from scratch during render. This takes FOREVER.

Solution:

1. **Click "Stop"** - Cancel the render
2. **Don't use Deliver page** - It triggers re-tracking
3. **Use Render in Place instead** (see Quick Fix section above)
4. Render in Place is more reliable at using cached tracking data

Why Render in Place works when Deliver doesn't:

- Render in Place uses the render cache more directly
- Deliver page sometimes re-computes effects even when cached
- It's a known quirk with Magic Mask specifically

PROBLEM: Render in Place ALSO Stuck on Tracking

If Render in Place is also re-tracking (slow progress):

Your cache is truly lost. Options:

1. **Wait it out** - Let it re-track (Render in Place is still faster than Deliver)
2. **If tracking is corrupted:** You may need to re-track manually, then Render in Place immediately after

To prevent this in future: Always Render in Place RIGHT AFTER tracking completes, before doing anything else.

GPU Acceleration - Is It Enabled?

If renders are extremely slow or stuck, check GPU settings:

```
DaVinci Resolve → Preferences → System → Memory and GPU
```

Setting	What to Check
GPU Processing Mode	Should be: CUDA (NVIDIA), OpenCL (AMD), or Metal (Mac) - NOT "Auto" if having issues
GPU Selection	Make sure your dedicated GPU is selected, not integrated graphics

Magic Mask Re-Computation During Render

Important: When you export from Deliver page, Resolve may **re-compute the Magic Mask** for every frame, even if you already tracked it.

Why?

- Render Cache ≠ Final Render
- The tracking DATA is cached, but the mask computation might still run during export

Symptoms:

- Render started but no progress for 10+ minutes
- Render is absurdly slow compared to similar projects
- GPU usage is maxed out

Solutions:

1. **Use Render Cached Images:**
   - Deliver page → Video tab → Advanced
   - ✅ Check "Use render cached images"
   - This uses pre-computed cache instead of recalculating
2. **Pre-cache before rendering:**
   - Playback → Render Cache → Smart
   - Wait for entire timeline to turn BLUE (fully cached)

- Then render with "Use render cached images" ON

3. **Render in Place first (best solution):**
    - Render in Place bakes the Magic Mask computation
    - Final exports then use the pre-baked file
    - No Magic Mask computation during final render

Render Stuck With No Progress?

Try this diagnostic:

1. Set In/Out points around just 10 seconds of footage
2. Try to render only that section
3. If 10 seconds takes forever, the problem is Magic Mask computation
4. If 10 seconds renders quickly, the problem might be a specific section of your timeline

Slow ProRes Renders?

ProRes itself encodes fast. If ProRes renders are slow:

- Magic Mask is being recomputed (see above)
- Check GPU acceleration is enabled
- Check cache location is on a fast SSD
- Try: Playback → Delete Render Cache → All, then re-cache, then render

---

FINAL TRUTH

A good grade is not noticed.
A bad grade is unforgettable.

Build reality first.
Then bend it gently.